AUDITORY SCIENCE MEETING 2026

8-9th September 2026

Core Technology Facility, Manchester, M13 9WU

The University of Manchester & University of Salford

Auditory Science Meeting (ASM) 2026

Programme & Abstracts

Tuesday 8 - Wednesday 9 September 2026

29 Talks	**1** Ted Evans Lecture	**48** Posters

Programme

Tuesday 8 September 2026

Time	Activity
09:15-09:45	Arrival, registration & poster set up
09:45-10:00	Welcome & housekeeping

Session 1: Speech communication, speech-in-noise and listening effort

ID	Time	Title	Speaker detail
O1	10:00-10:15	Effects of hearing loss and speech intelligibility on turn taking processes	Lauren V Hadley University of Nottingham
O2	10:15-10:30	Oculomotor tracking of selective attention in speech-in-noise listening	Xena Liu UCL Ear Institute
O3	10:30-10:45	Speak louder or turn down the noise? Comparing the processing demands of speech-in-noise versus quiet speech	Ronan McGarrigle University of Leeds
O4	10:45-11:00	Beyond intelligibility: listening effort, comprehension and enhancement of suprathreshold broadcast dialogue	William D. Curry University of Salford

Time	Activity
11:00-11:30	Break & poster viewing

Session 2: Cochlear implants and electric hearing

ID	Time	Title	Speaker detail
O5	11:30-11:45	Effects of charge interactions on perception of spectral shape by cochlear-implant listeners	Coline Marché University of Cambridge
O6	11:45-12:00	Vowel encoding and perception in acoustic and electric hearing	Andrew J. Oxenham University of Minnesota / University College London
O7	12:00-12:15	Using unmasking to measure the magnitude and effective polarity of cochlear-implant excitation patterns	Robert P. Carlyon University of Cambridge
O8	12:15-12:30	Can stimulus sampling based on temporal masking improve speech perception and power savings with cochlear implants?	Lidea Shahidi University of Cambridge
O9	12:30-12:45	The interaction between spectral degradation and cognitive effort during speech-in-noise perception with cochlear implants	Tobias Goehring University of Cambridge / University of Zurich
O10	12:45-13:00	Participatory design: involving families and clinicians in the little listeners study of neural processing of language in toddlers with bilateral cochlear implants (CI)	Sheila Flanagan University of Cambridge

Time	Activity
13:00-14:30	Lunch & poster viewing

Session 3: Auditory perception, attention, prediction and cognition

ID	Time	Title	Speaker detail
O11	14:30-14:45	Frequency discrimination at very low frequencies for sinewaves and complex sounds	Brian C. J. Moore University of Cambridge
O12	14:45-15:00	Confidence in auditory predictions influences brain and pupil responses: an EEG and eye tracking experiment	Buse Adam University College London
O13	15:00-15:15	Does the cocktail party effect affect partial loudness?	Josef Schlittenlacher University College London
O14	15:15-15:30	Native tone language experience can improve the ability to segregate voices by fundamental frequency	Stuart Rosen University College London
O15	15:30-15:45	Reduced pupil-linked arousal marks implicit memory for complex auditory sequences	Maria Chait UCL Ear Institute

Time	Activity
15:45-16:15	Break & poster viewing

The Ted Evans Lecture

ID	Time	Title	Speaker detail
T	16:15-17:15	How does auditory cortex construct auditory space?	Jennifer Bizley UCL Ear Institute

Speaker biography: Jennifer Bizley is a Professor of Auditory Neuroscience at the UCL Ear Institute and a Wellcome Career Development Fellowship holder. Her lab was established at UCL in 2011, and focusses on how the brain reconstructs auditory scenes, and how vision and movement may support listening in complex environments. Her lab's research combines behavioural testing in humans, with behaviour, electrophysiology and computational approaches in freely moving ferrets, and has been funded by the Wellcome Trust, Royal Society, BBSRC, RNID and the ERC.

Time	Activity
17:15-17:55	Poster viewing & networking
	ASM business meeting
19:15-22:00	Dinner at HOME

Wednesday 9 September 2026

Time	Activity
09:30-09:50	Arrival
09:50-10:00	Welcome & housekeeping

Session 4: Hearing-health risk, noise exposure and real-world listening

ID	Time	Title	Speaker details
O16	10:00-10:15	Auditory function following sport-related concussion and repetitive head impacts: a scoping review	Ewan Dean Lancaster University
O17	10:15-10:30	Investigating the impact of former contact sports participation on auditory function in older adults	Helen E. Nuttall Lancaster University
O18	10:30-10:45	Subclinical effects of recreational noise exposure: a pre-registered longitudinal study	Chris Plack The University of Manchester
O19	10:45-11:00	Using structural MRI to investigate the cortical effects of noise exposure	Sofya Koryagina University of Nottingham
O20	11:00-11:15	The Cadenza live concert experiment with Manchester Camerata	Rebecca R. Vos University of Salford

Time	Activity
11:15-11:45	Break & poster viewing

Session 5: Wider health factors, comorbidities and interdisciplinary translation

ID	Time	Title	Speaker details
O21	11:45-12:00	Hearing impairment and genetic risk for psychiatric disease converge in increasing auditory cortical synchronization in mice	Chen Lu UCL Ear Institute
O22	12:00-12:15	Auditory function differences between women and men: Is there a role for the female sex hormones?	Karolina Kluk The University of Manchester
O23	12:15-12:30	How does diabetes mellitus affect audiovestibular function? A thematic review of aetiological theories and controversies	Emma Stapleton The University of Manchester
O24	12:30-12:45	The roles of gut health and lifestyle factors in speech perception ability: a population-based comparison of older adults from the UK and the USA	Nez Sharp Lancaster University
O25	12:45-13:00	AURORA^3: a national facility for next-generation interdisciplinary research in audio, acoustics, and hearing science	Christos Chousidis University of Surrey

Time	Activity
13:00-14:30	Lunch & poster viewing

Session 6: Clinical rehabilitation, intervention and outcomes

ID	Time	Title	Speaker details
O26	14:30-14:45	Brain stimulation for hearing aid efficacy in age-related hearing loss	Jessica L. Pepper Lancaster University
O27	14:45-15:00	Auditory training for adults with hearing loss: Toward evidence-based practice	Natalie L. Lerigo-Smith NIHR Nottingham Biomedical Research Centre / University of Nottingham
O28	15:00-15:15	Do bone conduction hearing instruments alleviate listening effort in patients with single-sided deafness?	Claudia Contadini-Wright UCL Ear Institute
O29	15:15-15:30	Patient-defined success following stapes surgery for otosclerosis: a qualitative study highlighting divergence from conventional outcome reporting	Emma Stapleton The University of Manchester / Manchester Royal Infirmary

Time	Activity
15:30-16:30	Poster viewing & networking
16:30-16:45	Closing remarks

Oral Abstracts

O1. Effects of hearing loss and speech intelligibility on turn taking processes

Hadley LV

Hearing Sciences - Scottish Section, University of Nottingham, Glasgow, UK

Hearing loss affects almost 1/3 of adults over 60, with degraded speech input leading to fewer resources being available for higher level speech processing. I will present two studies investigating how hearing loss and speech intelligibility affect two critical turn-taking processes: response planning, and prediction of the end of a spoken turn. In a study of response planning, we demonstrate that at equivalent levels of intelligibility, older adults with normal and impaired hearing plan verbal responses equally early. A follow-up study investigated both response planning and prediction of a talker's turn end, testing three groups of older participants: normal hearing, hearing impaired listening at low demand, and hearing-impaired listening at high demand (but with high intelligibility). We found that all listeners were able to generate predictions of a turn end. However, only the hearing loss group that were listening easily (i.e., listening to louder stimuli) were able to use these predictions to produce verbal responses quickly; those with hearing loss listening with high demand showed delayed verbal responses. Together this work suggests that some issues relating to hearing loss in conversation may be a consequence of intelligibility, but others may be a cognitive consequence of hearing loss.

O2. Oculomotor tracking of selective attention in speech-in-noise listening

Liu X, Chait M

UCL Ear Institute, University College London, London, UK

Successful interaction with complex auditory scenes requires dynamic engagement of various cognitive processes, including arousal, attention, distractor suppression, and memory. Previous research has established that such information can be uncovered from our eyes, through orienting ocular responses like phasic Pupil Dilation (PD), Microsaccade rate(MS), and blinks. In the present experiment, we show that combining these measures provides rich information about time-to-time cognitive mechanisms in speech-in-noise processing.

We simultaneously recorded pupil and oculomotor data with eye-tracking during a speech-in-noise working memory task (N=40). The task required participants to attend to a target speaker while ignoring a temporally interleaved distractor speaker, all embedded in multi-speaker babble. Increased pupil responses (diameter and dilation rate) indicated heightened arousal in anticipation of and following target words, during which blink rates were reduced, suggesting also enhanced effort and attentional engagement. MSs were inhibited following specific time points of temporal attention, suggesting a cross-modal interaction, where visual sampling is reduced when attentional resources are deployed for audition. Frequency domain analyses also revealed robust ocular tracking of top-down attentional dynamics.

Importantly, time-locked analyses revealed signatures of attentional states at timepoints of stimulus processing ultimately associated with different behavioral outcomes. Compared to hits, miss events showed attenuated PD and oculomotor inhibition signals, indicating less successful attentional engagement, while enhanced ocular signals were observed in false alarms (with similar morphology to hits), signaling “incorrect” allocation of attention to distractors.

Overall, these results suggest that ocular dynamics can track time-to-time arousal and attentional engagement in listening, even in difficult listening conditions.

O3. Speak louder or turn down the noise? Comparing the processing demands of speech-in-noise versus quiet speech

McGarrigle R (1), Zheng R (2), Brims E (2), Knight S (3), Mattys S (2)

1. University of Leeds, Leeds, UK
2. University of York, York, UK
3. Newcastle University, Newcastle, UK

Both very quiet speech and speech in noise present challenges for the listener. Understanding quiet speech is demanding due to the intrinsic weakness of the signal, which, at very low levels, can be further interfered with by internal noise. In contrast, speech in noise involves energetic masking at all signal-to-noise ratios (SNRs) and might require additional cognitive resources to discriminate the target speech from background noise. In this study, we examined whether understanding speech in noise is more effortful than understanding quiet speech.

Participants were asked to repeat back sentences played with or without a speech-shaped noise masker (Noise vs. Quiet) while having their task-evoked pupil response (TEPR) tracked as a measure of listening effort. Each participant first went through a staircase procedure to determine either the SNRs (Noise condition) or the dB SPL thresholds (Quiet condition) corresponding to 90% and 50% accuracy. Participants then completed the main listening task, in which they were again asked to repeat back sentences presented at these individually tailored intensity levels. The design, therefore, involved Listening Condition (Noise vs. Quiet) and Difficulty Level (90% vs. 50%). Subjective effort ratings were also collected for each of these four conditions.

As expected, TEPRs were larger, and subjective ratings of effort higher, in the 50% than 90% condition. However, Listening Condition did not significantly influence either TEPRs or the subjective ratings of effort. No interaction was found between Listening Condition and Difficulty Level on any of the outcome measures. These results suggest that understanding speech in noise is not more inherently effortful than understanding quiet speech. Findings will be discussed in relation to current theoretical frameworks of everyday speech perception and effortful listening.

O4. Beyond intelligibility: listening effort, comprehension and enhancement of suprathreshold broadcast dialogue

Curry WC (1), Shirley BG (1), Cox TJ (1), Walsh M (2), Laverty T (2)

1. University of Salford, Manchester, UK
2. XPERI Inc., San Jose, CA, USA

Difficulty understanding dialogue remains a common broadcast complaint, even when content follows production guidance. Background music, effects and programme sound can increase effort, disrupt language processing and reduce memory for speech despite high intelligibility. Source separation can boost dialogue post-production. However, if content is already intelligible, is a level boost the best solution, or can targeted enhancement of speech-features reduce effort while preserving the intended mix?

We investigated this across two studies using broadcast-style audiobook material mixed with competing audio at broadcast-relevant SNRs. In a laboratory EEG experiment, acoustic challenge varied across three SNRs. After two-minute clips, listeners rated the listening experience across five subjective scales and completed a recognition task assessing the surface form, propositional memory and situation model. EEG was analysed using temporal response functions (TRFs) to examine links between neural speech tracking and participant measures.

Greater acoustic challenge produced more negative ratings and reduced recognition memory, with significant effects for surface-form and propositional information and reduced discrimination of contextually incorrect content. TRFs showed significant links between electrophysiological responses and participant measures, especially subjective ratings and acoustic-onset TRFs.

A subsequent online study tested whether enhancement improved these outcomes. Compared with an unprocessed baseline, both a 6 dB broadband boost and modulation-spectrum enhancement improved subjective ratings. Only modulation-spectrum enhancement significantly improved recognition memory, increasing surface-form accuracy relative to the control.

These findings show that comprehension can be reduced at moderate, broadcast-relevant SNRs, and indicate that at suprathreshold levels, effort and encoding depend not only on audibility, but also on temporal speech features.

O5. Effects of charge interactions on perception of spectral shape by cochlear-implant listeners

Marché C, Shahidi L, Guérit F, Carlyon RP

Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK

Modern cochlear-implant (CI) strategies employ interleaved pulsatile stimulation to minimise the between-channel charge interactions that can occur with simultaneous stimulation and that could distort the representation of spectral shape and potentially degrade speech perception. However, as pulse rates have increased, the interval between pulses on different channels has shortened. As a result, charge integration of non-simultaneous pulses at the auditory-nerve membrane may partially undo the benefits of interleaving.

We investigated the potential effects of non-simultaneous charge interactions on spectral-shape perception. Experiment 1 was based on the finding by Todd & Landsberger (JASA EL, 2018) who presented 16-channel interleaved pulse trains to listeners of the Advanced Bionics CI, and reported that the most-comfortable level could be reduced by an average of 3.6 dB by inverting the polarity of the pulses on alternate channels. We loudness-balanced these fixed (F)- and alternating (A) pulse trains for different numbers of electrodes: the effect remained substantial even with small subsets. We presented 7 Advanced Bionics listeners with a task in which each trial consisted of two pairs of stimuli, one of which differed in polarity (e.g. F-A vs F-F, A-F vs A-A), and asked them to report which pair sounded most different. Only 1 of the 7 listeners consistently chose the different-polarity pair. Hence for this stimulus charge interactions produced no effect on spectral shape perception. Participants were however very good at detecting a spectral tilt induced by one half of the array being alternating and the other fixed (els.9 to 16 alternating vs els.1 to 8 alternating). We are now extending this paradigm to study effects of charge interactions on the perception of spectral humps and notches, and to compare effects of interleaved vs simultaneous stimulation on spectral shape perception.

Todd & Landsberger, J Acoust Soc Am, 2018. DOI: 10.1121/1.5049701

O6. Vowel encoding and perception in acoustic and electric hearing

Oxenham AJ (1,2), Maxwell BN (1), Palatnik B (1), Milani M (1)

1. University of Minnesota, Minneapolis, MN, USA
2. University College London, London, UK

A classical question in auditory and speech perception relates to how vowels are encoded in the auditory periphery. Traditionally, vowels have been thought to be represented via their spectral profile, encoded by the distribution of auditory-nerve firing rate along the tonotopic axis (rate-place code). Alternatively, vowel formants may be encoded via a reduction in neural fluctuations in firing rate in neurons tuned to frequencies near the formants, relative to those observed in auditory-nerve fibers tuned between formants. We tested these alternatives in normal-hearing listeners and cochlear-implant (CI) users, using vowels processed to reduce access to either rate-place or fluctuation cues.

Brief steady-state vowels were passed through a tone-excited envelope vocoder and processed to reduce either rate-level or fluctuation cues by equating the amplitudes of all the tone carriers or by removing amplitude fluctuations from the carriers, respectively. Both CI listeners and normal-hearing listeners were tested in their ability to discriminate the vowels. The CI listeners were additionally tested in their ability to distinguish between vocoded and unprocessed stimuli.

Vowel identification was substantially better in conditions that maintained rate-place cues than in conditions that maintained only fluctuation cues, where performance was near chance for both CI and normal-hearing listeners. In addition, most CI listeners found it difficult to distinguish the vocoded from the unprocessed stimuli, suggesting that they were perceptually equivalent.

Overall, the results suggest that both CI and normal-hearing listeners use rate-place, rather than fluctuation-based, cues to identify steady-state vowels in isolation. The fact that tone vocoding was perceptually transparent for most CI users supports the use of tone vocoding when seeking to dissociate competing cues via vocoded stimuli. [Supported by NIH grant R01 DC023255.]

O7. Using unmasking to measure the magnitude and effective polarity of cochlear-implant excitation patterns

Carlyon RP, Guerit F, Deeks JM

Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK

Methods that aim to provide more focused stimulation by cochlear-implant (CI) electrodes inject current whose magnitude and polarity vary along the electrode away. We describe and validate a new psychophysical technique that measures both magnitude and, importantly, polarity of stimulation along the neural excitation pattern, thereby allowing evaluation and refinement of focused-stimulation methods and with the aim of improving speech perception.
We presented users of the Advanced Bionics CI with a continuous baseline stimulus consisting of a 480-pulse-per-second (pps) train of symmetric biphasic pulses whose phases were in anodic-cathodic (AC) or cathodic-anodic (CA) order. Participants detected a 50-ms 480-pps train of probe pulses, each consisting of a short anodic phase followed by a cathodic phase of 1/8th the amplitude and 8 times longer. Each probe pulse immediately followed a baseline pulse. A comfortable-level MP baseline with pulses in CA order reduced thresholds for probes on the baseline electrode to 15 dB below those in quiet. We attribute this substantial unmasking to charge summation between adjacent anodic phases of baseline and probe pulses. Unmasking decreased with decreasing baseline level and with increasing separation between the baseline and probe electrodes, thereby providing a measure of tuning. No unmasking was observed with baseline pulses in AC order. When we presented the baselines in “wide tripolar” mode (central electrode number = 8, opposite-polarity flankers [4,12] or [2,14]) in CA and AC conditions, and measured thresholds for probes on each stimulating electrode, unmasking was observed only at electrodes and conditions where the second baseline phase was anodic. This finding reveals reversals in the effective polarity of stimulation across the excitation pattern of focused stimuli. We are using this method to refine novel stimulation methods to sharpen excitation patterns, uncontaminated by unwanted “side-lobes” of excitation.

O8. Can stimulus sampling based on temporal masking improve speech perception and power savings with cochlear implants?

Shahidi L (1), Carlyon RP (1), Vickers DA (2), Goehring T (1,3)

1. Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK
2. Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK
3. Department of Otorhinolaryngology, Head and Neck Surgery, University Hospital Zurich, University of Zurich, Zurich, Switzerland

New developments in cochlear implants (CIs), including power-intensive noise-reduction methods and fully implantable devices, require minimising the amount of power used to stimulate electrodes, which additionally benefits battery life. By using a model of temporal masking to sample the CI stimulation pattern, the temporal integrator processing strategy (TIPS) reduces the number of pulses presented, improves power, and maintains or enhances speech perception (Lamping et al. 2020; Shahidi et al. 2026). Here, we determine whether TIPS provides perceptual or power-savings benefits compared to a uniform reduction in pulse rate, as employed clinically. Two double-blind within-subject experiments tested 12 CI recipients. Experiment 1 measured speech perception as a function of uniform rates of 80 to 900 pulses per second (pps). Experiment 2 compares TIPS against a low uniform rate, setting TIPS parameters to yield stimuli with the same average number of stimulation pulses. In both experiments, stimuli were assessed using behavioural measures of consonant discrimination, sentence intelligibility, and pitch ranking and compared against the listener's clinically used rate. Experiment 1 indicated that speech perception was surprisingly robust to reductions in pulse rate down to 240 pps, below which it deteriorated. Experiment 2 is ongoing and will indicate whether TIPS better maintains cues important for consonant, sentence, and pitch perception, than its uniform-rate counterpart. Speech perception is robust to reductions in channel stimulation rate well below those used clinically, suggesting low uniform rates could be a site of potential power savings. Results from Experiment 2 will indicate whether further power savings benefits are possible with TIPS by allowing further reduction in the number of stimulation pulses without impacting speech perception.

Lamping et al., Hear Res, 2020. DOI: 10.1016/j.heares.2020.107969
Shahidi et al., Ear Hear, 2026. DOI: 10.1097/AUD.0000000000001741

O9. The interaction between spectral degradation and cognitive effort during speech-in-noise perception with cochlear implants

Goehring T (1,2), Dang TN (1), Shahidi L (1), Carlyon RP (1)

1. Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK
2. Department of Otorhinolaryngology, Head and Neck Surgery, University Hospital Zurich, University of Zurich, Zurich, Switzerland

Electric current spread and unwanted interactions between spectral channels are key factors that limit the perception of speech for cochlear implant (CI) recipients. CI speech perception likely requires high cognitive effort to work out what has been said, especially in noisy situations. However, the relationship between signal resolution and cognitive demand is not well understood. We studied the causal interaction between signal resolution and cognitive effort during CI speech perception in noise. First, we used spectral blurring to causally manipulate signal resolution and combined this with (1) a novel speech recognition task to manipulate cognitive effort and (2) a manipulation of neural spread of excitation. Two groups performed a within-subject study: 12 normal-hearing (NH) adults listening to vocoder-based CI simulations and 12 adult CI recipients. There were highly significant effects of spectral blurring on speech-in-noise perception across groups and participants. The cognitive manipulation significantly affected speech recognition scores in noise for both groups. However, only for the CI group there was a significant interaction effect between spectral blurring and cognitive effort. The CI group showed a change in susceptibility to higher spectral blurring amounts under increased cognitive effort, which supports the assumption that signal resolution and cognitive demand influence each other in a complex manner. The neural spread manipulation did not lead to a change in performance and there was no interaction, suggesting that this manipulation was not effective. These findings support the assumption that multiple factors combine and interact to limit speech perception success with CIs, including background noise, signal degradations and cognitive effort. The differences between individual listeners and between the two groups indicate varying degrees of resilience to the different sources of perceptual difficulty or the ability to compensate for them.

O10. Participatory design: involving families and clinicians in the little listeners study of neural processing of language in toddlers with bilateral cochlear implants (CI)

Flanagan SA, Vickers DA, Billing A

University of Cambridge, Cambridge, UK

Many toddlers benefit greatly from cochlear implantation prior to age 12 months. However, receptive and spoken language outcomes can vary widely between children. Addressing these individual differences is exacerbated by the lack of objective tools available to clinicians during critical periods of auditory and language development. Standard audiological assessment relies on behavioural responses requiring a maturity that is not yet developed in young children. To address this, our study's aim is to develop a clinical tool using High-Density Diffuse Optical Tomography (HD-DOT) to objectively assess cortical markers of language development in paediatric CI recipients whilst listening to and viewing speech in noise.

Patient and Public Involvement and Engagement (PPIE) has emerged as a cornerstone in health and social care research, ensuring that the voices of patients, service users, and carers are integrated at every stage of the research process. This approach fosters transparency, trust, and mutual learning between researchers and the communities they serve. The benefits of embedding PPIE are well known, including improved study design, participant recruitment and retention, and the generation of outcomes that better reflect the needs and priorities of end users.

Here we implemented a three-stage action research study design. This involved consultations with i) experienced parent groups of CI children, ii) clinician groups with a representative mix of professions (e.g. Paediatric Audiologist, Speech and Language Therapist) and iii) newer families of CI recipients aged up to 2 years. All meetings were conducted onsite in our research rooms to demonstrate equipment. Reflecting on our feedback will allow us to adapt our protocol to improve recruitment and retention of the families in the study. By identifying the obstacles to participation, we aim to develop an essential, objective approach for monitoring speech acquisition in infants with CIs.

O11. Frequency discrimination at very low frequencies for sinewaves and complex sounds

Jurado C (1), Moore BCJ (2)

1. Norwegian University of Science and Technology, Trondheim, Norway
2. University of Cambridge, Cambridge, UK

Frequency discrimination measured as a function of frequency has often been used to assess the lower limit for pitch (LLP). Usually, the stimuli have been harmonic complex tones with spectral content above 100 Hz. Here, frequency difference limens (FDLs) were measured for three stimulus types: sinusoids (pure tones, denoted PT), harmonic complex tones containing harmonics 1-3 (HT), and “transposed tones” (TT) created by modulating a 125-Hz carrier with a half-wave rectified sinusoidal modulator. The TT were designed to give a single sharp envelope peak per period, thereby enhancing frequency discrimination based on a temporal mechanism. The repetition frequency, f, of the stimuli varied from 12 Hz to 48 Hz. All stimuli were presented at a loudness level of about 60 phons, to reduce loudness cues. Frequency discrimination was measured using a two-alternative forced-choice task. At the end of each run, subjects were asked to judge whether the sounds were “Discontinuous/vibration like” or “Continuous/tone-like”.

The FDLs increased progressively as f was decreased, to more than 10% of f at f = 12 Hz, but did not show a break point corresponding to the LLP. This may reflect the use of cues other than pitch at very low f (e.g. judging the rapidity of the perceived fluctuations). For f = 17 and 24 Hz, the FDLs were significantly larger for the HT than for the PT. The percentage of “tone-like” judgements decreased with decreasing f and was less than 50% for f = 12, 17 and 24 Hz. For f from 24 to 48 Hz, the TT were judged as more tone-like than the PT, while for f = 12 and 17 Hz the percentage of tone-like judgements was low for all stimulus types. The pattern of the results was well explained by a model based on the summary autocorrelation function (SACF).

O12. Confidence in auditory predictions influences brain and pupil responses: an EEG and eye tracking experiment

Adam B, Chait M

University College London, London, UK

The brain is increasingly viewed as a prediction system that learns patterns to predict future events. This is vital for everyday life to recognise speech or detect changes in sound. A key question is whether the brain's response to a change is shaped by confidence in the preceding sound pattern. To investigate, we ran an EEG-pupillometry experiment, n=40. EEG measured neural activity; pupil size provided an index of tonic (sustained) and phasic (event-related) arousal. Participants listened to rapid tone sequences where the statistical structure changed midway through. Results show that both brain and pupil responses track changes in sequences, and pupil responses reveal how changes in auditory structure influence arousal-related systems. Next, we used computational models to estimate the confidence in the sound pattern before a change, reflecting the reliability of predictions in upcoming sounds. We then compared responses following relatively high/low confidence. Results showed that neural responses were shaped by the predictability of the context prior to a change. Transitions following high confidence contexts elicited later neural responses, suggesting violations of reliable predictions are processed differently in brain. Pupil responses affected by confidence. Before transition, high confidence sequences evoked smaller sustained pupil responses, revealing that predictable contexts require less processing effort. After transition, high/low confidence contexts showed different response patterns, suggesting distinct mechanisms underlying the processing of new events. Notably, these effects occurred with identical stimuli, revealing the differences in neural and pupil responses are not due to the sounds themselves. Instead, they reflect how the auditory system interprets sounds in light of the preceding context. Overall, this highlights a mechanism crucial to auditory scene analysis where context influences perception, attention, and responses to auditory events.

O13. Does the cocktail party effect affect partial loudness?

Schlittenlacher J (1), Lee C (1), Foottit E (1), Moore BCJ (2)

1. University College London, London, UK
2. University of Cambridge, Cambridge, UK

Humans with normal hearing have a remarkable ability to focus on a single talker in a mixture of competing voices, effectively suppressing irrelevant speech, a phenomenon called the cocktail party effect. Spatial separation of the target and interfering talkers can improve speech reception thresholds by 7 dB or more. In the present study, we investigated whether this benefit extends beyond speech intelligibility to partial loudness, that is the perceived loudness of a target sound in the presence of background noise.

In a series of experiments with headphones using artificial-head recordings, we established the level difference required for equal loudness (LDEL) of the target speech when the target talker was collocated with the competing speech or was spatially separated from it. Tested spatial separations were 22.5, 45 and 90 degrees, with the target fixed at locations of 0° and 45° and the competing speech varied in position. LDELs were obtained using a 1-up/1-down procedure and four tracks per condition.

With a single competing talker and reference signal-to-noise ratio (SNR) of 0 dB, all LDELs were close to 0 dB when the target was in front (0°), regardless of spatial separation. When the target was located at 45°, LDELs increased with increasing spatial separation up to 1.5 dB. With four competing talkers, all from the same direction, and the target at 45°, LDELs were up to 2.5 dB for an SNR of 0 dB and up to 3.6 dB for an SNR of -6 dB.

Overall, the effect of spatial separation on loudness was smaller than for speech intelligibility. In contrast to speech intelligibility, loudness judgments depend primarily on the temporal portions with highest loudness, not the entire sound. For a single competing talker, judgments may have been based on portions when the short-term SNR was highest, leading to a smaller effect of spatial separation. For four competing talkers, the short-term SNR varied less over time.

Supported by the Royal Society, grant 232164.

O14. Native tone language experience can improve the ability to segregate voices by fundamental frequency

Rosen S, Solomatina M

University College London, London, UK

It is often reported that native speakers of tone languages are more sensitive to pitch changes than native speakers of non-tonal languages, even when the pitch contrasts are not linguistic. We assessed how native linguistic experience (tonal vs. non-tonal language background) and musical aptitude influence the use of voice pitch (F0) cues for talker segregation in multitalker environments. The listening task used matrix sentences (from one male and one female talker) of the form ‘Show the [animal] where the [colour] [number] is’. Target sentences used ‘dog’ as the animal, whereas masker sentences were of the same form, with no overlap in animal, colour, or number.

The listening task involved identifying the colour and number of the target in the presence of two simultaneous masking sentences (at identical RMS levels) from the same talker as the target. Speech reception thresholds (SRTs) were determined adaptively. Masker sentences were presented with their original F0 values and with pitch contours shifted by 4 and 8 semitones (shifted downward for the female talker and upward for the male). Musical abilities were assessed using an online music aptitude test. Thirty-seven non-native, UK-resident L2 English speakers (18 tonal, 19 non-tonal) participated.

Results show that increasing F0 separation significantly improved SRTs for both listener groups. Crucially, tonal language speakers derived greater benefit from F0 differences than non-tonal speakers, reflected in a slightly greater release from masking as the pitch difference increased from 0 to 8 semitones (5.9 dB vs. 4.8 dB). In contrast, musical aptitude did not significantly predict performance. Overall, these findings suggest that tonal language experience is linked to a somewhat more effective use of pitch cues for speech segregation, highlighting the role of long-term auditory experience in shaping the ability to perceive speech in the presence of competing speech.

O15. Reduced pupil-linked arousal marks implicit memory for complex auditory sequences

Leung WS (1), Yeung L (1), Huviyetli M (1,2), Bianco R (3), Chait M (1)

1. UCL Ear Institute, University College London, London, UK
2. Faculty of Health Sciences, Izmir Bakircay University, Izmir, Türkiye
3. Department of Translational Research on New Technologies in Medicine and Surgery, University of Pisa, Pisa, Italy

Observers rapidly form long-lasting implicit memories for recurring sensory patterns, supporting adaptive behaviour through learning of environmental contingencies. Here we tested how such memories interact with neuromodulatory systems that regulate arousal—whether implicit familiarity increases arousal through retrieval-related processes or instead decreases arousal via more efficient sensory processing.

We examined these competing accounts using complex auditory tone-pip sequences in which a subset of patterns recurred approximately every three minutes. One group of participants (N=30) actively detected transitions from random to regular structure, whereas a second group (N=40) heard the same sequences while performing a decoy task. Consistent with prior work, recurring regular sequences elicited faster reaction times in the active group, confirming the formation of implicit memory.

Subsequently, both groups listened to previously encountered regular patterns (REGr) and matched novel regular patterns (REGn) while pupil diameter was recorded. Across groups, REGr sequences evoked significantly smaller pupil dilations than REGn sequences, indicating attenuated arousal for familiar patterns. This reduction was sustained until sequence offset in the active group but was transient in the exposure group, suggesting that task demands during exposure shape the stability of pupil-linked memory expression. Importantly, the REGr–REGn pupil difference did not correlate with explicit post-session recognition, demonstrating that the effect was not driven by conscious awareness.

Together, these findings identify pupil dynamics as a robust physiological marker of learned auditory structure and show that implicit auditory sequence memory attenuates arousal. The results support predictive processing accounts in which familiarity reduces uncertainty and, consequently, neuromodulatory drive.

O16. Auditory function following sport-related concussion and repetitive head impacts: a scoping review

Dean E (1,2), Andrew J (1,2), Oyewole J (1,2), Gaffney C (2), Slade K (2), Nuttall H (1)

1. Department of Psychology, Lancaster University, Lancaster, UK
2. Lancaster Medical School, Lancaster University, Lancaster, UK

Head impacts are common in sport, and auditory symptoms, like difficulty understanding speech-in-noise, are frequent. Yet auditory assessment remains absent from concussion protocols. Repetitive head impacts are linked to neurodegenerative diseases, highlighting the need for sensitive, accessible biomarkers. Auditory processing relies on precisely timed subcortical and cortical neural transmission, making it vulnerable to diffuse axonal injury and the neurometabolic cascade of concussion. Further, the auditory cortex lies within the lateral sulcus, a region highly vulnerable to concussive impacts. This vulnerability suggests auditory measures may provide a window into concussion-related neuropathology. To explore how sports-related head impacts affect auditory function, a scoping review was conducted. Eligible studies assessed auditory function in athletes following concussive, sub-concussive, or repetitive head impacts in sport. Thirty-one studies met criteria. Peripheral hearing remained within clinically normal limits, yet central auditory processing deficits emerged within hours to years' post-concussion. Subcortically, speech-evoked frequency following responses showed reduced fundamental frequency amplitude. Cortically, N100 amplitude reductions suggest early disruption of auditory processing. Higher-order auditory function, measured with auditory oddball paradigms, demonstrated reduced P300 amplitude, even after symptoms had resolved. Contact sport participation was also associated with poorer speech-in-noise performance and elevated rates of tinnitus and noise sensitivity. Despite converging evidence, auditory assessment remains absent from concussion management. While gaps remain, including limited data on female, elite, and retired athletes, the auditory system represents an underutilised biomarker of concussion-related neuropathology with implications for monitoring brain health and protecting athletes' perceptual and communication abilities.

O17. Investigating the impact of former contact sports participation on auditory function in older adults

Andrew JR (1,4), Thomas A (2), Plack CJ (3), Gaffney CJ (4), Nuttall HE (1)

1. Department of Psychology, Lancaster University, Lancaster, UK
2. Brainbox Ltd, Cardiff, UK
3. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
4. Lancaster Medical School, Lancaster University, Lancaster, UK

Sports-related head impacts in younger adult contact sports athletes are associated with disrupted auditory cortical responses (Andrew et al., 2026). It is unclear whether such auditory disruption persists into later life. This study investigated the impact of contact sports on auditory function in older adults who were formerly engaged in contact sports.

Data were collected from former contact (n=16) and control (n=13) participants, who were recreationally active (data are preliminary from an overall planned sample of 70). Participants were matched across cognitive status and average pure-tone audiometry thresholds. Auditory N100s to /da/ in quiet and multi-talker babble conditions were recorded, as well as the signal-to-noise ratio (SNR) associated with participants' 80%, 50%, and 20% accuracy performance points during a sentence-in-noise task. Parietal alpha power was also measured during the sentence task to index listening effort.

For N100 data, condition was significant ($F(1,26) = 8.81$, $p = .006$), with reduced N100 amplitudes in background noise ($M = -3.30$ μV, $SD = 1.77$) compared to quiet ($M = -4.90$ μV, $SD = 2.62$). Neither group nor the interaction were significant. For the behavioural data, performance level was significant ($F(2,54) = 728.99$, $p < .001$), and a group x performance interaction ($F(2,54) = 3.81$, $p = .028$). The interaction reflected that the contact group required a more favourable signal-to-noise ratio to achieve 80% accuracy ($M = 2.74$ dB SNR, $SD = 1.98$), relative to controls ($M = 1.28$ dB SNR, $SD = 1.30$). Alpha power analyses revealed no significant findings at this stage.

These preliminary findings suggest that understanding speech in background noise is more challenging for former contact sports athletes relative to older adult control participants, even with clinically normal peripheral hearing and cognitive status. Neural data are underpowered at this stage, and further data collection is ongoing with results to be discussed.

Andrew et al., 2026. DOI: 10.64898/2026.02.13.705730

O18. Subclinical effects of recreational noise exposure: a pre-registered longitudinal study

Guest H (1), Dewey R (2), Kluk K (1), Millman R (1), Munro K (1), Murray C (1), Prendergast G (1), Plack C (1)

1. The University of Manchester, Manchester, UK
2. University of Nottingham, Nottingham, UK

Inconsistent evidence for subclinical auditory effects of noise exposure may reflect prior studies' limitations (small samples, retrospective exposure estimates, cross-sectional designs). This prospective study aimed to overcome these issues with a large cohort, repeated measures, and pre-registered hypotheses.
220 adolescents with normal hearing (16–17 at baseline) completed a ~3-hour audiological test battery at baseline and ~3 years later; 199 (90%) returned for follow-up and 191 (87%) maintained normal hearing. The battery measured extended high-frequency thresholds, distortion product otoacoustic emissions (DPOAEs), auditory brainstem response (ABR) wave I amplitude, and middle ear muscle reflex; noise exposure was self-reported at 18 months. Pre-registered primary analyses tested for longitudinal associations between noise exposure and changes in these four physiological outcomes. Exploratory analyses considered speech perception, tinnitus, demographic differences (sex, skin tone), lifestyle factors (nicotine, alcohol, deprivation), and relationships among measured variables. The protocol was pre-registered (OSF DOI: 10.17605/OSF.IO/GHD97).
Median 3-year noise exposure was ~36 concert equivalents (two-hour, 100 dBA), with a 95% range of 0.5–690. Despite wide exposure variability, no significant association was found between noise exposure and changes in any primary outcome measure. Exploratory analyses also found no noise-related changes in speech perception, tinnitus, or other measured outcomes.
Adolescents with widely varying recreational noise exposure over three years showed no evidence of permanent clinical or subclinical auditory damage. Typical exposures in this age group may be less harmful than assumed, or any adverse effects might only appear later with ageing. The full dataset will be openly shared for further collaborative analyses.

O19. Using structural MRI to investigate the cortical effects of noise exposure

Koryagina S (1,2,3), Dewey RS (1,2,3), Asghar M (1), Guest H (4,5), Millman R (4,5), Prendergast G (4,5), Sereda M (2,3), Plack CJ (4,5), Francis ST (1)

1. Sir Peter Mansfield Imaging Centre, School of Physics and Astronomy, University of Nottingham, Nottingham, UK
2. NIHR Nottingham Biomedical Research Centre, Nottingham, UK
3. Hearing Sciences, School of Medicine, University of Nottingham, Nottingham, UK
4. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
5. NIHR Manchester Biomedical Research Centre, Manchester, UK

Noise-induced hearing loss (NIHL) is common and preventable. Standard pure-tone audiometry only detects NIHL after perceptible hearing loss occurs, missing earlier, subclinical damage. The aim of the study was to identify early biomarkers of noise-induced damage using behavioural, audiological and MRI measures.
200 participants were grouped into younger adults (G1), normal-hearing older adults with low (G2) vs high (G3) noise exposure, and older adults with NIHL (G4). Participants completed extended high frequency (EHF) audiometry, distortion product otoacoustic emissions (DPOAEs), Digits-in-Noise (DiN) and MRI1. Here we focus on structural MRI. T1-weighted MRI scans were processed using FreeSurfer7.4.1 for cortical reconstruction and volumetric segmentation. Group-level statistical analysis was conducted using FreeSurfer's general linear model (GLM) framework to assess cortical thickness (CT) and grey matter volume (GMV). Further, we examined analysis in 74 FreeSurfer ROIs to quantify region-specific CT and GMV. Analyses tested the effects of age, noise exposure and NIHL.
EHF audiometry and DPOAEs showed significant ($p<0.001$ *Bonferroni corrected) group differences with age and NIHL, and multi-talker babble DiN with NIHL. There was no effect of noise exposure. Voxel-wise analysis of MRI group maps demonstrated significant widespread changes with age, and changes in areas of medial temporal gyrus and frontal cortex related to the combination of high noise exposure with and without NIHL. ANOVAs showed significant group effects in 21 GMV ROIs, including bilateral superior and middle temporal gyri ($p<0.001$*), and 31 CT ROIs, including left Heschl's gyrus ($p<0.001$*), driven by age. No significant effects were related to noise exposure alone.
EHF audiometry, DPOAE and DiN differentiate the effect of NIHL, while MRI predominantly differentiates age effects. No effects of noise exposure alone were seen.

Dewey et al., Aperture Neuro, 2025. DOI: 10.52294/001c.140461

O20. The cadenza live concert experiment with Manchester Camerata

Vos RR (1), Akeroyd MA (2), Bannister S (3,4), Barker J (5), Cox TJ (1), Roa Dabike G (5), Fazenda B (1), Firth J (2), Graetzer S (1), Greasley A (3), Whitmer WM (2)

1. Acoustics Research Centre, University of Salford, Manchester, UK
2. Hearing Sciences, School of Medicine, University of Nottingham, Nottingham, UK
3. School of Music, University of Leeds, Leeds, UK
4. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
5. Department of Computer Science, University of Sheffield, Sheffield, UK

Research on music listeners with hearing loss such as the “Hearing Aids for Music” (HAfM) project, has identified challenges including reduced dynamic range, distortion, and poor sound quality. However, few studies have examined these experiences in live orchestral settings. This study will present the Cadenza Live Concert Experiment, an ecologically valid investigation of music perception in hearing aid users, conducted in collaboration with Manchester Camerata at Stoller Hall, at 3pm on Thursday, the 29th October, 2026.

Participants with hearing loss will attend a live concert featuring Romantic repertoire (e.g., Beethoven Symphony No. 6) alongside a structured experimental component. During the performance, the orchestra will present controlled musical excerpts in which parameters such as dynamics and orchestral balance are systematically varied. Audience members will complete in-situ questionnaires capturing demographic and hearing-related information, alongside perceptual ratings of audio quality, loudness, clarity, and distortion. Open-ended responses will also be collected to document subjective listening issues during full movements.

The study will aim to identify musical and acoustic conditions that contribute to degraded listening experiences for hearing aid users in real-world contexts. By integrating perceptual ratings with spatial data (seat location) and performance variables, the experiment will generate new insights into how hearing aid processing affects the live music listening experience.

This work will demonstrate the value of concert-based research for capturing realistic listening experiences and advancing inclusive musical engagement and participation for listeners with hearing loss. Findings will inform the design of future studies and lead to improvements in hearing aid signal processing and fitting strategies for both live and recorded music, and improve accessibility of concert hall design.

Cadenza Challenge. https://cadenzachallenge.org

Hearing Aid for Music National Survey. https://musicandhearingaids.org/2016/09/22/hafm-national-survey

O21. Hearing impairment and genetic risk for psychiatric disease converge in increasing auditory cortical synchronization in mice

Lu C (1), Linden JF (1,2)

1. UCL Ear Institute, University College London, London, UK
2. Department of Neuroscience, Physiology and Pharmacology, University College London, London, UK

Both hearing impairment and genetic risk for psychosis are hypothesized to shift the cortical excitation-inhibition balance toward excitation, yet the definition, nature and impact of this shift remain debated. The Df1/+ mouse model of human 22q11.2 Deletion Syndrome carries a homologue of a genetic risk factor for psychosis, and also exhibits high rates of conductive hearing impairment like human carriers (Fuchs et al., 2013, PLoS ONE). Auditory evoked potentials (AEPs) in Df1/+ mice show abnormal growth with increasing sound intensity and inter-stimulus interval (Lu and Linden, 2025, Translational Psychiatry), suggesting increased auditory cortical excitability. Here, we ask whether hearing impairment and the 22q11.2 deletion have similar or distinct effects on local field potentials (LFPs) and single-unit activity in the auditory cortex.

To mimic hearing impairment observed in a subset of the Df1/+ mice (n=6 hearing impaired, n=5 normal hearing), we performed malleus removal surgery on WT mice at P11 (n=11 malleus removed, n=6 sham). Both single-unit activity and LFPs were recorded using Neuropixels 1.0 probes in awake head-fixed mice, during presentations of 16kHz tone pip trains with varying inter-stimulus intervals (200-1000 ms) and relative sound levels (threshold+0-30dB).

We found increased LFP power in both the hearing-impaired groups (Df1/+ and WT mice) and Df1/+ normal hearing group, particularly in the beta and gamma band. Consistent with previous AEP results, these animal groups also showed increased level- and interval- dependence of LFP power and greater synchronization of single-unit activity.

Together, these findings demonstrate that both hearing impairment and the 22q11.2 deletion shift auditory cortical dynamics toward increased excitation and enhanced neural synchronization. Thus, two distinct risk factors for psychosis in humans converge in producing similar auditory cortical dysfunction.

Fuchs et al., PLoS ONE, 2013. DOI: 10.1371/journal.pone.0080104
Lu & Linden, Transl Psychiatry, 2025. DOI: 10.1038/s41398-024-03218-x

O22. Auditory function differences between women and men: Is there a role for the female sex hormones?

Aloufi N, Marschall K, Heinrich A, Kluk K

The University of Manchester, Manchester, UK

The potential modulatory role of female sex hormones in hearing remains unclear. We investigated whether short-term (menstrual cycle) and long-term (menopause) changes in hormone levels influence hearing sensitivity, intensity discrimination, and speech-in-noise performance.

We conducted a dual-phase, repeated-measures observational study in hospital audiology clinics, including 55 participants: 20 pre-menopausal women (mean age ~22 years), 15 age-matched men (~22), and 21 post-menopausal women (~53), all with normal hearing. Each participant attended five weekly test sessions; serum oestradiol and progesterone levels were measured each session. Audiological measures included pure-tone audiometry (PTA), extended high-frequency (EHF) audiometry, just-noticeable differences for intensity (JNDI), and speech perception in noise using the International Speech Test Signal (ISTS) with a matched long-term average speech spectrum (LTASS) noise.

Pre-menopausal women showed systematic auditory fluctuations across menstrual phases: hearing thresholds were more sensitive and intensity discrimination improved in the late follicular (oestradiol-dominant) phase compared to other phases. Men and post-menopausal women showed no comparable changes. Post-menopausal women had overall poorer hearing sensitivity than pre-menopausal women, reflecting a long-term hormonal effect. Speech perception in noise was not significantly influenced by hormone fluctuations.

Female sex hormone fluctuations, especially elevated oestradiol, can transiently enhance auditory sensitivity and intensity discrimination in pre-menopausal women, while persistently lower hormone levels after menopause are associated with reduced hearing sensitivity. Considering hormonal status in audiological assessments may help optimise hearing care for women.

O23. How does diabetes mellitus affect audiovestibular function? A thematic review of aetiological theories and controversies

Stapleton E, Kluk K, Plack CJ

The University of Manchester, Manchester, UK

A connection between diabetes and audiovestibular deficits has long been acknowledged, yet definitive evidence of an aetiological link remains elusive. This review aims to address recurring controversies in the literature, in the context of study design.

Following the Synthesis Without Meta-analyis (SWiM) guidelines, a comprehensive literature review was performed, filtering 119 manuscripts into the final analysis. Included manuscripts underwent qualitative analysis using a grounded theory approach; extracted data was used to compile a summary of aetiological theories and recurring controversies. These were critically analysed in the context of study design.

Proposed aetiological theories for effects of diabetes on audiovestibular function include a variety of possible lesions, across the entire auditory pathway. Recurring controversies include whether the effect is real; which hearing frequencies are affected; whether severity or duration of diabetes plays a role, and whether diabetic microangiopathy is the primary aetiology.

Future clinical research requires meticulous matching of cases and controls; sensitive biomarkers for the detection of early damage; consideration of, and control for, the effects of acute blood glucose levels; longitudinal studies to examine effect time; sensitive measures of blood glucose fluctuation; and careful consideration to the potential impact of confounding factors. Definitive correlation between clinical research findings, and cellular-level changes, will also be informative.

O24. The roles of gut health and lifestyle factors in speech perception ability: a population-based comparison of older adults from the UK and the USA

Sharp I (1), Guest H (2), Peelle JE (3), Plack CJ (2), Slade K (1), Nuttall HE (1)

1. Lancaster University, Lancaster, UK
2. The University of Manchester, Manchester, UK
3. Northeastern University, Boston, MA, USA

Speech perception ability is a key metric of hearing health, reflecting the impact of hearing loss on daily function. 20% of the global population lives with hearing loss, but it does not affect all members of society equally. A cumulation of lifestyle factors impacts hearing health; yet lifestyle is impacted by factors difficult to modify, such as geographical location, cultural norms and socioeconomic status. Sociological and environmental differences between countries, such as the degree of cultural individualism, may interact with lifestyle factors to modify their impact on hearing function. This study investigated whether geographical location (UK vs USA) impacts the influence of lifestyle factors on speech perception ability in older adults. Insights from this research highlight the need for country and culture-specific approaches to preventing hearing loss. An online design collected data via Prolific (www.prolific.com) on demographics, geographical location, BMI, diet, and physical activity. Participants also completed a digits-in-noise task to assess speech perception ability. 848 adults aged 60-84 were recruited, with an equal split between countries. The mean age of UK and US participants was not significantly different, $t(846) = -1.42$, $p = .156$. The multiple regression model, modelling the effects of lifestyle factors on speech perception, was significant, R^2 adjusted =.08, $F(10, 797) = 7.95$, $p < .001$. There was a significant interaction of geographical location and physical activity on speech perception ($\beta = 0.61$, $p = .007$), and US citizens showed a trend between poorer speech perception and increased physical activity ($R = -0.1$, $p = .043$). Increasing age significantly predicted poorer speech perception ($\beta = 0.07$, $p < .001$). To further explore the results, research utilising clinical predictors (e.g. pure tone thresholds) should be used to supplement findings. America's highly individualised culture, which deprioritises public health, should be considered.

O25. AURORA³: a national facility for next-generation interdisciplinary research in audio, acoustics, and hearing science

De Sena E, Mason R, Chousidis C, Ali R, Liado Gonzalez P

University of Surrey, Guildford, UK

Emerging challenges in hearing science require integrating auditory perception, immersive sound, and realistic listening scenarios. Yet most auditory experiments rely on highly absorbent test spaces, unable to reproduce complex environments of everyday listening. Others take place in natural spaces, where acoustic conditions cannot be controlled or systematically varied. This limits repeatability and the ability to investigate complex interactions between listeners and their acoustic environment.

To address these challenges, the IoSR at the University of Surrey is developing AURORA³, a research facility designed to support accurate, repeatable, and highly controlled auditory experiments and data collection.

AURORA³ comprises two complementary environments, a full anechoic chamber equipped with a spherical loudspeaker array for sound-field reproduction and HRTF measurements and a passive Variable Acoustics Room (VAR), where reconfigurable acoustic panels enable precise control of reverberation. The VAR is the first facility of its kind to incorporate a movable wall system, offering emulation of different room volumes and acoustic interaction between adjacent spaces. Further capabilities include a modular loudspeaker system, optical motion capture and tracking, and robotic source–receiver positioning for investigating dynamic acoustic scenarios.

The facility will support a broad range of research, including speech perception in complex environments, hearing aid and assistive-listening device evaluation, spatial hearing and HRTF studies, auditory scene analysis, electrophysiological investigations, and virtual and augmented reality studies. Designed as a national resource, AURORA³ will provide academic and commercial users with flexible access, technical expertise and state-of-the-art equipment. It aims to accelerate interdisciplinary research and enable next-generation experiments in environments that combine experimental control with real-life listening conditions.

O26. Brain stimulation for hearing aid efficacy in age-related hearing loss

Pepper JL (1), Warren JD (2), Plack CJ (3), Slade K (1), Nuttall HE (1)

1. Lancaster University, Lancaster, UK
2. University College London, London, UK
3. The University of Manchester, Manchester, UK

By the age of 70, nearly 80% of people experience some degree of hearing loss. Reduced auditory signal can lead to atrophy of auditory brain regions. Whilst hearing aids may offset such neuroplastic changes, adjusting to hearing aids can be time-consuming. The neural and behavioural benefits of hearing aids may be realised more quickly by using facilitatory Transcranial Magnetic Stimulation (TMS) to support hearing aid adaption. This study investigated whether increasing the neuroplastic potential of the brain with TMS can accelerate the benefits of hearing aids for older adults with age-related hearing loss.

Preliminary data (N=14) are reported from participants aged 60-85 who received hearing aids in the last 3 months (target sample N=92). Participants were divided into active and sham TMS groups for a 5-day TMS intervention. Neural responses to sound (event-related potential: N100), audiological performance (QuickSIN) and cognitive function (memory, executive function, processing speed) were compared before (Timepoint 1) and after (Timepoint 2) the TMS intervention. Participants will also be measured 6 months later.

Data indicate that mean N100 amplitudes decrease from Timepoint 1 to Timepoint 2 in the active TMS group: -2.58 (SD=2.54) to -2.25 (SD=2.16), yet increase in the sham TMS group: -2.18 (SD=1.96) to -2.63 (SD=1.85). Mean QuickSIN scores were 2.74 (SD=1.66) and 2.74 (SD=1.82) at Timepoints 1 and 2 in the active group, compared with 3.40 (SD=2.66) and 3.17 (SD=2.13) in the sham group. Differences in N100 and QuickSIN are non-significant. A mixed ANOVA revealed a significant main effect of timepoint on cognition ($F(1,12)=19.82$, $p<.001$); across groups, cognitive performance was better at Timepoint 2 (M=113.79, SD=9.66) than Timepoint 1 (M=102.93, SD=13.11).

Current results indicate that cognitive scores may improve with hearing aid adaption. Data collection is ongoing, and we will observe how TMS benefits neural and behavioural responses to speech.

O27. Auditory training for adults with hearing loss: Toward evidence-based practice

Lerigo-Smith N (1,2), Zuk N (3), Heinrich A (4,5), Broome E (1,2), Akeroyd M (1,2), Henshaw H (1,2)

1. NIHR Nottingham Biomedical Research Centre, Nottingham, UK
2. Hearing Sciences, School of Medicine, University of Nottingham, Nottingham, UK
3. NTU Department of Psychology, Nottingham Trent University, Nottingham, UK
4. NIHR Manchester Biomedical Research Centre, Manchester, UK
5. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK

Adults with hearing loss frequently report difficulty understanding speech-in-noise, even when using hearing aids. Digital auditory training (AT) offers a potential intervention; however, despite robust pooled evidence of generalised benefits to speech perception, findings vary across studies. Further work is needed to understand how AT works, and how trained improvements transfer to improvements in untrained tasks for generalised real-world benefits. Practice Listening and Understanding Speech (PLUS) is a web-based auditory–cognitive training platform that enables the systematic manipulation of perceptual and cognitive training task demands, providing a practical opportunity to address this knowledge gap. Our research investigated whether manipulating these demands for two auditory-cognitive training tasks affects the magnitude of on-task learning and generalised improvements.

A remote online preregistered randomised controlled trial will recruit 120 adults (≥18 years) with an unaided SRT of -5.5 dB SNR or higher on the Digit Triplet Test. Participants are allocated to one of two training conditions defined by relatively greater or lesser auditory and cognitive task demands (easier vs harder) across two adaptive tasks; phonemic discrimination or competing speech training (~n=30 across each of the four training groups). Easy conditions involve immediate phoneme identification (0-back) or gender-separated target and masker speech, whereas hard conditions introduce working-memory demands (1-back) or same-gender competing speech.

Following the pre-training baseline assessment, participants train online for 30 minutes/day, 5 days/week, over two weeks (10 sessions; 5 hours total) at home on their own device. The primary outcome is speech-in-noise perception, using a modified Coordinate Response Measure. Secondary outcomes include trained-task performance, cognition, and self-reported listening, communication, and quality of life. Preliminary results will be presented.

O28. Do bone conduction hearing instruments alleviate listening effort in patients with single-sided deafness?

Contadini-Wright C (1), Anekpichayasit T (1), Fu X (1), Chung M (2), Mehta N (1,2), Chait M (1)

1. UCL Ear Institute, University College London, London, UK
2. Royal National ENT and Eastman Dental Hospitals, London, UK

Single-Sided Deafness (SSD) presents significant challenges in complex listening environments. Although individuals can rely on their functioning ear to meet everyday demands, doing so often incurs substantial cognitive costs, like increased listening effort, attentional demands, and fatigue. Bone Conduction Hearing Instruments (BCHIs) reroute sound to the better ear, but their ability to reduce these cognitive burdens remains unclear.

A spatially separated adaptation of the Coordinate Response Measure (CRM) task was administered to typical-hearing controls (N=29) and BCHI users with SSD (N=25). Participants completed High Load (occluded ear for controls; BCHI off for patients) and Low Load (unoccluded ears for controls; BCHI on for patients) conditions. Speech was presented to the deaf/occluded ear, while noise was presented to the hearing ear. Spatially separated control data were additionally compared with previously collected diotic data to validate the adapted paradigm.

Listening effort and attentional allocation were assessed using 3 oculomotor measures: Pupil Diameter (PD), indexing arousal; Microsaccade Rate (MSR). indexing attentional allocation; and Eye-Blink Rate (EBR), indexing cognitive processing demands. Although BCHI activation significantly improved behavioural performance, little evidence was found that it reduced arousal or attentional demands in challenging listening conditions. However, EBR analyses revealed load-dependent differences in post-keyword rebound dynamics, suggesting that BCHI use may facilitate shorter processing times and a faster release from processing.

These findings indicate that BCHIs may not alleviate all dimensions of listening effort, but may support more efficient computational processing in SSD. More broadly, they demonstrate the value of integrating non-invasive physiological measures into clinical practice to provide a more comprehensive understanding of the cognitive consequences of auditory interventions.

O29. Patient-defined success following stapes surgery for otosclerosis: a qualitative study highlighting divergence from conventional outcome reporting

Boorman D (1), Stapleton E (1,2)

1. The University of Manchester, Manchester, UK
2. Manchester Royal Infirmary, Manchester, UK

Outcome reporting in stapes surgery traditionally focuses on audiometric measures, with limited patient perspectives and no standardised definition of success. This may result in a mismatch between reported outcomes and themes that matter to patients.

A qualitative descriptive study was conducted using a structured questionnaire incorporating open-ended questions and Likert-scale items. The questionnaire was developed in collaboration with patient representatives and informed by existing outcome reporting in literature. Adult patients who had undergone stapes surgery for otosclerosis within the preceding 24 months, with minimum 12 months postoperative follow-up, at a tertiary centre were invited to participate. Free-text responses were analysed using inductive thematic analysis; descriptive quantitative data were used to contextualise findings.

22 of 28 invited patients responded (78%). Four overarching themes were identified: change in hearing, social impact, tinnitus, and surgical complications. Patients defined success as a multidimensional construct centred on functional hearing improvement, independence from hearing aids, and absence of complications. Improvement in tinnitus and quality of life were also important. Participants also highlighted outcomes that are not routinely reported in the literature, including the ability to function in real-world listening environments and the absence of serious complications.

There is a mismatch between outcomes reported in stapes surgery literature and those prioritised by patients, and between published definitions of success and patient-defined success. These findings provide patient-informed outcome domains that will support the development of a core outcome set for stapes surgery, enabling more meaningful and standardised outcome reporting.

The Ted Evans Lecture Abstract

T1. How does auditory cortex construct auditory space?

Bizley J

University College London, London, UK

Sound localisation is essential for survival and requires that the brain compute location from differences in the sound wave at the two ears. While dedicated brainstem nuclei extract sound localisation cues, auditory cortex is necessary for accurate sound localisation behaviour. However, it is not clear what value auditory cortex adds to the information extracted earlier in the auditory pathway. In this talk I will present our attempts to answer this question over a number of years, by performing neural recordings from animals engaged in tasks that are more complex and naturalistic than traditional approach-to-target localisation paradigms. For example, in the real world, both sounds and listeners move. In contrast, in the lab, we tend to fix listeners within a static speaker ring and present static isolated stimuli. By mapping spatial receptive fields in freely moving animals, we were able to demonstrate that auditory cortical neurons represent auditory space in both head and world-centered reference frames, and multiplex variables like movement speed into their responses. Ongoing work in the lab has explored how experience impacts the formation of world-centered spatial receptive fields. Most recently, to better mimic real-world localisation, we have built a large arena with an acoustically transparent floor under which an array of 48 speakers is located. This allows animals to 'hunt' sounds through the arena while we track their head movements in 3D and record neural activity with chronically implanted Neuropixels. We hope that these experiments will allow us to understand how active sensation through head movements supports sound localisation, and the role of auditory cortex in this.

Poster Abstracts

Poster quick list

ID	Presenter	Title
P1	Zheng Pan	Crossmodal plasticity effects of hearing loss and ageing in the human auditory cortex
P2	John Francis Culling	Modelling binaural unmasking at 125 Hz
P3	Brian Roberts	Effects of sudden correlated and anti-correlated level changes across tone subsets on the build-up of stream segregation
P4	Lorenzo Bonoldi	Improving music for those with hearing difference: a participatory approach using professional sound engineers
P5	Neil J Ingham	Auditory temporal responses of mutant mice and diagnosis of hearing impairment pathology
P6	Pedro Lladó	Perceptual models of spatial hearing for optimising loudspeaker reproduction
P7	Yimeng Li	Refining and piloting a parent- and teacher-behaviour intervention to support pragmatic development in deaf children
P8	El F Smith	Developing an inner speech battery for children
P9	Emma Foottit	Simple auditory reaction times as a tool to measure hearing
P10	Caoilan McConville	Investigating changes in temporal response function with increased difficulty of speech-in-noise tasks due to differing background noise volumes, types and an increased working memory load
P11	Ilaria Atzori	Investigating the utility of EEG power-bands to measure the involvement of auditory working memory in speech-in-noise perception
P12	Scott Bannister	Listening intentions of hearing aid users in everyday life: an ecological momentary assessment study
P13	Barbara Manini	Cross-modal representations in the auditory cortex of deaf and hearing individuals
P14	Sarah Knight	No evidence of a benefit for speech-in-noise intelligibility from recalibration to auditory-visual asynchrony
P15	Bindiya Patel	Does voice familiarisation training improve speech-in-speech intelligibility for older adults with presbycusis?
P16	Rutendo Chatiza	Towards modelling the impact of spread of excitation on classical psychophysical paradigms with cochlear implant users
P17	Nadia Aghili	Head-Centred Sound Localisation Behaviour Reveals Head and World Centered Reference Frames in Auditory Cortex
P18	Ronan McGarrigle	Language proficiency drives listening-related fatigue in those who communicate in their second language
P19	Megan Griffiths	The social and temporal influences on attention in audiovisual scenes

P20	Katharina Kaduk	Perceptual distinctiveness of place and voicing mispronunciations in hearing primary school aged children
P21	Natasha Winge	Within- and between-session variability in DPOAE measurements in an individual-differences study of auditory perception
P22	Clémence Hallou	Does optogenetics based inactivation of the secondary auditory cortex influence vowel discrimination in noise in ferrets?
P23	Simone Graetzer	The first Cadenza lyric intelligibility prediction challenge
P24	Adele Simon	Exploring noise sensitivity in autism through cortical speech tracking
P25	Arturas Verbickas	Perceiving the absent: auditory expectations generate illusory percepts under inattention
P26	Ifat Yasin	Association between the Envelope Following Response (EFR), loudness scaling, and speech perception
P27	Nicholas R. Haywood	Comparison of eCAP estimates of neural health in cochlear implant listeners
P28	Guido Sebastian Lax	Towards better cortical auditory evoked potential biomarkers for schizophrenia risk: investigations in a mouse model
P29	François Guérit	Current focussing with cochlear implants: on efficiently assessing and optimizing its impact
P30	Tobias Goehring	A web-based listening test system and its validation for remote testing and clinical practice with cochlear implant recipients
P31	Chi Chung Brian Lam	Sex differences in forward-masking ABR Wave I recovery coincide with reproductive transitional stages in CBA/Ca mice
P32	Gabriel Salazar	Assessing suitable fNIRS hyperscanning paradigms in a naturalistic classroom laboratory
P33	Brandon L. O'Hanlon	Design and validation of a synchronisation system for different sensors across time in a naturalistic classroom
P34	Christian J. Sumner	The challenge of quantifying audiovisual integration of speech
P35	Jakub T. Sztandera	Physiological markers of auditory situational awareness in complex spatialised scenes
P36	Jennifer Firth	Radian: Measuring the rate of change of speech-in-noise performance across one year
P37	Karolina Kluk	Measures of subclinical hearing function: long-term test-retest reliability and relations between change outcomes
P38	Jude L R Barker	Relating illusory tone perception to tinnitus
P39	Veronica M. Tarka	Pitch-based stream segregation of complex sounds in ferrets
P40	Michael A Stone	Exploring the contribution of spectrum and modulation in the EHF region to speech intelligibility
P41	William M. Whitmer	The effects of control and stimulus on the self-adjustment of frequency-dependent gain
P42	Stefan Bleeck	Towards ecologically valid measures of listening effort: a novel comprehension-based paradigm to assess cognitive load in noise

P43	Marina Salorio Corbetto	Developing a paradigm to assess the impact of cognitive status on cochlear-implant audiovisual benefit
P44	Garreth Prendergast	Exploring the link between hyperacusis, anxiety and depression in a non-clinical UK cohort
P45	Alistair Beith	Exploring the context-dependent nature of conversational success
P46	Debi Vickers	Factors influencing 10-year post implantation language skills in children with cochlear implants
P47	Zaal Contractor	Time course of reverberation adaptation when listening to natural sentences
P48	Karolina Kluk	The effect of hearing aids on cognitive resources and the ease and effectiveness of communication: a randomised controlled trial

P1. Crossmodal plasticity effects of hearing loss and ageing in the human auditory cortex

Pan Z (1), Połeć M (1), Avgerinos A (1), Bi K (1), Green T (2), Cardin V (1,*)

1. Deafness, Cognition and Language Research Centre, University College London, London, UK
2. Department of Speech, Hearing and Phonetic Sciences, University College London, London, UK

*Corresponding Author

Cross-modal plasticity refers to the phenomenon by which typical sensory areas respond to stimulation in other sensory modalities when their main input is absent or reduced. In this study, we examined the effect of age-related hearing loss (ARHL) on cross-modal plasticity in auditory brain regions. ARHL is prevalent in a large proportion of older adults, and it is associated with an increased risk of cognitive decline and dementia. Previous research in congenitally deaf adults has shown that cross-modal effects are task-dependent, with stronger responses for conditions that require higher executive processing. Here, we conducted an fMRI experiment in older adults to determine whether task-dependent cross-modal recruitment of auditory cortical regions also occurs following ARHL. In the MRI scanner, participants with and without ARHL completed a visual task switching paradigm and a visual working memory task, each comprising high and low executive demand conditions. Results from both groups of participants showed that the high demand conditions reliably activated canonical frontoparietal regions involved in executive function and cognitive control. Crucially, we also observed significant recruitment of auditory regions during these visual tasks, particularly under higher executive demands. Cross-modal activations in auditory areas occurred in both groups, with no significant effects of hearing level or age. These findings indicate that auditory regions are involved in high executive demand visual processing, suggesting that task-dependent cross-modal plasticity is not restricted to early sensitive periods. We propose that reduced auditory input and age-related changes in cortical processing jointly contribute to the reorganisation of the auditory cortex in later life. Understanding these mechanisms is critical for clarifying the neural consequences of ARHL and their impact on cognitive decline.

P2. Modelling binaural unmasking at 125 Hz

Culling JF

Cardiff University, Cardiff, UK

At low frequency, the difference in binaural masking level difference (BMLD) between N0Sπ and NπS0 for broadband noise maskers increases. According to equalization-cancellation theory, this increase occurs because, within an auditory filter, the π phase difference in the noise at each ear corresponds to a widening range of time delays across its bandwidth. This wide range of delays cannot be equalized so well with a single compensating time delay, resulting in only partial cancellation. However, the predicted difference is smaller than observed empirically, suggesting a secondary causation factor. The predicted difference can be enlarged by assuming wider auditory filters, less effective processing for larger compensating delays (a "delay cost"), or by a hard limit on the available compensating delays. The modelling presented here shows that the predictions under these different possibilities diverge using unusually low signal frequencies, such as 167 Hz, combined large delays to the noise (Nτ). For a 1-ms hard limit, the BMLD for NτS0 is very low for τ ≥ 2 ms, but not for the other two. For a doubled bandwidth, the BMLD for NτS0 is nearly 15 dB for τ = 3 ms and falls steeply for τ ≥ 4 ms, but not for the other two. For a delay cost, BMLD for NτS0 is 5-7 dB at 1-3 ms and declines gradually with further increases τ. This delay-cost pattern is the most similar to the only existing data at low frequency (Jeffress et al. 1962, J. Acoust. Soc. Am. 34, 1124-1126).

Jeffress et al, J Acoust Soc Am, 1962. DOI: 10.1121/1.1918262

P3. Effects of sudden correlated and anti-correlated level changes across tone subsets on the build-up of stream segregation

Rajasingam SL (1,2), Summers RJ (1), Roberts B (1)

1. Aston University, Birmingham, UK
2. Anglia Ruskin University, Cambridge, UK

The tendency to hear ABA– tone sequences as segregated builds up over time but can be reset by sudden changes in tone properties. Two experiments explored the effects of correlations in level change (6-dB differences) across A and B tone subsets. 12 listeners heard 20-s-long sequences (tones=100 ms; base frequency=1 kHz; Δf=4-8 semitones). They continuously monitored each sequence and reported when they heard the sequence as integrated or segregated. Experiment 1 used triplets in a LHL– (low-high-low-silence) configuration to compare the effects of correlated and anti-correlated level changes (i.e., one rises when the other falls) every 4 s in the A and B subsets. In control conditions where level was fixed throughout, the percept was reported as more integrated when both subsets had the same level than when they differed by 6 dB. Correlated rises in level for the A and B subsets resulted in resetting, but correlated falls in level had little effect. Less resetting was evident in the anti-correlated cases and occurred only for transitions involving a rise in B-tone level and a fall in A-tone level (B↑/A↓-transitions). For small Δfs, the opposite configuration of level changes (A↑/B↓-transitions) tended to elicit greater segregation than for the corresponding times in the fixed-difference sequences (i.e., overshoot). Experiment 2 compared the effects of anti-correlated level changes in LHL– and HLH– triplet contexts to identify whether the directional effects of transitions in the A and B subsets on reported streaming arose from their relative frequency or within-triplet position. Again, B↑/A↓-transitions caused resetting; A↑/B↓-transitions increased segregation, which for the HLH– sequences led to marked overshoot. Overall, HLH– sequences usually induced more segregation than LHL– sequences. The extent of stream segregation depends not only on the correlation of sudden changes in level between tone subsets but also on the within-triplet positions of these tones.

P4. Improving music for those with hearing difference: a participatory approach using professional sound engineers

Bonoldi L (1), Cox TJ (1), Hart AM (2)

1. Acoustics Research Centre, University of Salford, Manchester, UK
2. Salford School of Arts, Media and Creative Technology, University of Salford, Manchester, UK

This study investigates how audiometric diversity shapes professional music mixing practices, challenging the assumption that audio production is optimised for a single, normal-hearing listener. Using a participatory mixed-methods approach, hearing-diverse engineers are engaged as co-investigators to explore how perceptual differences influence technical decision-making.

The study combines interviews, audiometric assessment, remix tasks, and signal-based analysis. A descriptor–metric mapping framework (DMMF) links perceptual descriptors (e.g. clarity, brightness, openness) with measurable features such as spectral centroid, low-mid energy (250–500 Hz), crest factor, loudness (LUFS), and stereo width.

Findings show consistent adaptive strategies grounded in individual auditory profiles. Perceptual improvements are not achieved through increased loudness, stereo width, or high-frequency enhancement. Instead, clarity and listening comfort emerge through redistribution of spectral energy, reduction of low-mid masking, preservation of dynamic contrast, and controlled spatial organisation. Many hearing-optimised mixes present lower loudness and narrower stereo images while achieving greater clarity and stability.

The study also highlights the role of perceptually adaptive tools. Visual and interactive interfaces act as perceptual supports, helping users identify masking, spectral balance, dynamics, and spatial distribution that may not be fully audible. By enabling user-driven adjustment and making parameters perceptually transparent, these tools support inclusive, perception-aware workflows.

This work reframes hearing-diverse engineers as perceptual experts and positions mix quality as relational and context-dependent. Future work extends into HCI through the co-design and evaluation of adaptive mixing interfaces, exploring how interactive tools can enhance clarity, reduce listening effort, and support personalised control across diverse listening profiles.

P5. Auditory temporal responses of mutant mice and diagnosis of hearing impairment pathology

Ingham NJ (1), Plack CJ (2), Steel KP (1)

1. Wolfson Sensory, Pain and Regeneration Centre, King's College London, London, UK
2. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK

Using mutant mice, we are developing physiological tests that may be transferrable into the clinic to diagnose hearing impairment pathology. These mice carry different single gene mutations and have known sites of lesion and dysfunction, covering the three major categories of sensorineural hearing impairment described by Schuknecht & Gacek (1993); Sensory (hair cell dysfunction), Metabolic (stria vascularis dysfunction), and Neural (auditory neuron defects).

Amplitude Modulation Following Responses (AMFR) were evoked by 12 or 24 kHz carrier tones presented at 80 dB SPL (100 ms duration, 5 ms rise/fall time), 100 % sinusoidally amplitude-modulated at rates (modulation frequency, fm) from 100-1750 Hz. We calculated AMFR magnitude and synchrony coefficient as a function of fm.

AMFR amplitude vs fm transfer functions were multi-peaked in normal hearing control mice, having peaks close to modulation rates of 300 Hz, 600 Hz and 1000 Hz. Klhl18<lowf> (IHC dysfunction), Mir96<tm3.1Wtsi> (IHC & OHC dysfunction), Slc26a5<tm1-Cre> (Prestin, OHC dysfunction), and Wbp2<tm2a> (dysfunctional innervation) mutant mice showed reductions in AMFR amplitude and synchrony at these three fm ranges. Sgms1<tm1b> and S1pr2<stdf> mutant mice (stria vascularis dysfunction, with reduced endocochlear potential) showed reduced AMFR amplitudes and synchrony in the 300 Hz and 600 Hz ranges, but in contrast to the other mutants examined, these AMFRs were almost indistinguishable from littermate controls at the higher modulation rates tested.

Reduced AMFR amplitudes in hearing-impaired mice may be a general feature across "neural" and "sensory" pathologies, in contrast to the normal AMFR amplitudes seen at higher modulation rates in "metabolic" impairment. These observations may contribute to a diagnostic test battery to differentiate the sites of pathology of inner ear sensorineural hearing loss and help to inform better treatment strategies.

Schuknecht & Gacek, Ann Otol Rhinol Laryngol, 1993. DOI: 10.1177/00034894931020S101

P6. Perceptual models of spatial hearing for optimising loudspeaker reproduction

Lladó P, Daugintis R, Souchaud AR, De Sena E

University of Surrey, Guildford, UK

Loudspeaker-based audio reproduction has traditionally been optimised using physical measures of the sound field rather than perceptual criteria. In this project, we collect listening-test data for loudspeaker reproduction to evaluate the performance of existing auditory models of spatial hearing. After validating these models against the perceptual data, we use them as virtual listeners to optimise loudspeaker driving signals for desired spatial attributes within the listening area. We introduce POLAR, an end-to-end differentiable framework for optimising loudspeaker signals using perceptual loss functions. POLAR enables joint optimisation over multiple perceptual attributes and multiple listener positions, providing a flexible and perceptually grounded method for spatial audio reproduction with loudspeakers.

P7. Refining and piloting a parent- and teacher-behaviour intervention to support pragmatic development in deaf children

Li Y, Chilton H, Coupe N, Heinrich A

The University of Manchester, Manchester, UK

Pragmatics is the social use of language. Children primarily acquire pragmatics through incidental learning, which can be enhanced by rich communicative environments and opportunities created by caregivers, including parents and teachers. Deaf and Hard of Hearing (DHH) children often experience delayed pragmatics, resulting in delays in social participation and lifetime achievements. While several interventions aim to support pragmatic development in DHH children either directly or via their parents, evidence- and theory-based interventions with clear mechanism-driven change and proven efficacy are yet to be developed.

The overarching aim of the PhD project is to develop such an intervention by adapting the following existing resource aimed at parents and Teachers of the Deaf (ToD): Supporting the Pragmatic Skills and Social Communication Skills of Deaf Children.

This study follows the MRC framework. Step 1, evidence profile generation. All suggested communicative behaviours currently included in the resource are reviewed regarding their efficacy, theoretical mechanisms, and implementation issues in enhancing pragmatics in DHH children. Step 2, resource refinement. Focus group sessions are conducted with parents and ToDs. Their perspectives on resource acceptability and barriers and facilitators to communicative behaviour enactment are then combined with the evidence profiles generated in Step 1 to inform resource refinement. Step 3, intervention package development. Remaining behaviour change support gaps are identified in the refined resource, and the intervention package is developed by retaining existing behaviour change support and addressing identified gaps. Step 4, intervention testing and refinement. Stakeholder workshops, Patient and Public Involvement and Engagement, and a pilot study are conducted to test and refine the intervention.

The results of steps 1 and 2 will be presented in the poster.

P8. Developing an inner speech battery for children

Smith EF, Yao B, Stewart HJ

Lancaster University, Lancaster, UK

Most adults and children speak to themselves, when this is done internally it is called inner speech (Baars, 2003; Flavel et al., 1997). Vygotsky (1987) proposed that inner speech develops through the internalisation of social and private speech. This suggests that language ability is central to individual differences in inner speech. Inner speech has been linked to several cognitive behaviours such as arithmetic (Holland and Low, 2010), emotional recognition (Camminga et al., 2025) and executive function (Baron and Ardel, 2022). Inner speech is broadly researched in adults, using methods from self-report to linguistics assessments. Many of these assessments are not suitable be used with children due to developmental appropriateness. In children, inner speech is typically studied using task-suppression paradigms. A decreased performance under articulatory suppression indicates greater reliance on inner speech during cognitive tasks. Although the suppression method supports the impact of inner speech in cognitive tasks, this method does not directly study inner speech. We tested whether child-based language assessments predict stronger inner speech reliance during cognitive tasks, with the aim of validating these assessments as indicators of inner speech ability. We will be presenting our data so far from our target of 42 neurotypical hearing children, aged 7-11. Each child will complete: three cognitive assessments under motor and articulatory suppression; four language assessments; and hearing and listening assessments.

Components from a principal components analysis on the language assessments will be entered into three separate linear regression models, one for each of the cognitive task difference score. Significant predictors will indicate that the language assessments successfully reflect the children's inner speech ability. This will provide a more direct methodology for our future studies exploring the impact of noise on children's inner speech.

Baars, J Conscious Stud, 2003.
Flavel et al., Child Dev, 1997. DOI: 10.2307/1131923
Vygotsky, The collected works of L. S. Vygotsky, 1987. DOI: 10.1007/978-1-4613-1655-8
Holland & Low, Br J Dev Psychol, 2010. DOI: 10.1348/026151009X424088
Camminga et al., Cogn Psychol, 2025. DOI: 10.1080/20445911.2025.2478887
Baron & Ardel, Perspect ASHA Spec Interest Groups, 2022. DOI: 10.1044/2022_PERSP-22-00042

P9. Simple auditory reaction times as a tool to measure hearing

Foottit E, Schlittenlacher J

University College London, London, UK

Simple auditory reaction times (RTs) have the potential to serve as an objective measure for evaluating hearing ability. Previous research has demonstrated that RTs vary systematically with loudness and frequency. The present study expands upon this work by investigating RTs in 20 normal-hearing young adults across a wide range of frequencies and sound pressure levels, at 125, 250, 500, 1000, 2000, 4000 and 8000 Hz and from 40 to 70 dB in 5 dB increments. Each condition was repeated 36 times per participant.

Preliminary findings (n = 10) indicate that both frequency and loudness influence RTs. Faster responses were generally associated with higher levels, while RTs varied across frequencies. Notably, a dip in RTs was observed at 4000 Hz, consistent with patterns predicted by equal-loudness contours, which reflect the heightened sensitivity of human hearing in this frequency region. These findings support the use of RTs as a sensitive behavioural measure of auditory perception and hearing function.

P10. Investigating changes in temporal response function with increased difficulty of speech-in-noise tasks due to differing background noise volumes, types and an increased working memory load

McConville C, Atzori I, Heinrich A, Perugia E

The University of Manchester, Manchester, UK

The ability to hear and understand speech in noisy environments is moderated by working memory capacity (Rudner et al., 2011). Much of the previous research has focused on the correlation of working memory and intelligibility on speech-in-noise tasks. This study goes further by investigating the electrophysiological correlates of working memory and speech-in-noise perception, by means of EEG. Normal-hearing, native English-speaking adults aged 18-35 took part in the investigation. Participants were asked to listen to sets of short, sentences while continuous EEG signal was collected from 32 electrodes. The sentences were presented in blocks of 4 or 6 sentences, in quiet, in a background of babble or in white background noise, under two conditions, active and a passive. In the passive condition participants simply listened to the sentences under each type of background noise. The active condition, participants had to recall the last word of each sentence once the block was complete. The outcome measure was to determine how the changes in background noise type and volume level affect the Temporal Response Function (TRF). TRF is autocorrelation that expresses the similarity between the speech envelope to and the overall envelope response of the EEG. Data collection is ongoing. It is hypothesised that increased working memory capacity will cause a decrease in similarity between speech and EEG envelopes. Hence, the increased number of sentences presented will increase working memory capacity and reduce the TRF. Compared to white noise, babble will cause an increase in working memory capacity therefore possibly decreasing TRF. The changes in signal-to-noise ratio with background levels may cause either increased or decreased WM engagement depending on overall intelligibility of speech (Mattys et al, 2025).

Rudner et al., J Am Acad Audiol, 2011. DOI: 10.3766/jaaa.22.3.4

P11. Investigating the utility of EEG power-bands to measure the involvement of auditory working memory in speech-in-noise perception

Atzori I, Perugia E, Heinrich A,

The University of Manchester, Manchester, UK

Working memory (WM), a cognitive system that temporarily holds and manipulates information necessary for complex tasks (Baddeley, 1992), is a crucial element for speech-in-noise (SiN) perception (Hjortkjær et al., 2020). Auditory cortical activity synchronises to intelligibility as it relates to acoustic speech features (Hjortkjær et al., 2020). However, it is unknown whether auditory cortical activity also reflects WM load influences, and which power-band most accurately indicates WM engagement in SiN perception. The present study investigates this question for speech presented under different listening conditions. Electroencephalography was used to record brain activity while 15 18–35-year-old, native English speakers listened to sentences presented in three background conditions: quiet, babble, and white noise, in blocks of either 4 or 6 sentences. The sentences in the shorter blocks were also presented at two different signal-to-noise ratios (SNR). For each background noise condition, participants completed passive and active conditions, with the latter requiring participants to recall the final word of each sentence. These two manipulations allowed us to examine how changes in listening difficulty and WM load influence neural activity. Neural activity was measured via alpha (8-12 Hz) and theta (4-8 Hz) band power, which are sensitive to WM processing (Yurgil et al., 2020). Data collection is ongoing. It is hypothesised that more challenging listening conditions (babble and white noise), a lower SNR, and increased WM load will require greater cognitive resources and produce greater changes in neural oscillatory activity associated with speech processing. Theta power is expected to increase with WM load (Raghavachari et al., 2001), reflecting greater demands on maintenance, manipulation, and retrieval of information (Hsieh & Ranganath, 2014). Alpha power is also expected to increase, reflecting heightened executive demands (Luo et al., 2005).

Baddeley, Science, 1992. DOI: 10.1126/science.1736359
Hjortkjær et al, Eur J Neurosci, 2020. DOI: 10.1111/ejn.13855
Yurgil et al., Front Psychol, 2020. DOI: 10.3389/fpsyg.2020.00266
Raghavachari et al., J Neurosci, 2001. DOI: 10.1523/JNEUROSCI.21-09-03175.2001
Hsieh & Ranganath, Neuroimage, 2014. DOI: 10.1016/j.neuroimage.2013.08.003
Luo et al., Neuroimage, 2005. DOI: 10.1016/j.neuroimage.2005.05.040

P12. Listening Intentions of Hearing Aid Users in Everyday Life: An Ecological Momentary Assessment Study

Bannister S (1), Christensen JH (2), Whiston H (1), Saunders G (1)

1. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
2. Eriksholm Research Centre, Snekkersten, Denmark

Hearing aid (HA) satisfaction often depends on combinations of listening conditions (e.g., listening environment, noise) and listening intentions of the user. Listening conditions can be precisely determined and HAs can automatically adjust to optimise for these. However, HAs cannot do this for listening intentions, resulting in limited personalisation of sound processing, and a mismatch between HA settings informed by controlled experiments and ecologically valid listening intentions. This ecological momentary assessment study aims to understand dynamic listening intentions of HA users in everyday life.

Bilateral HA users with varied degrees of hearing loss will, for two days within a seven-day period, continuously report their listening intentions by responding to surveys every 15 minutes for 14 hours. In each survey, participants report whether their listening intentions have changed since the previous prompt; if so, they will describe their intentions (e.g., what they are doing; what they are trying to listen to; what else they can hear) using open-ended text or voice recordings. They then rate on a Visual Analog Scale: 1) importance of being able to hear what they are trying to hear; 2) how negatively or positively they feel about their hearing ability right now; and 3) how difficult it is to hear what they are trying to hear. Finally, participants can upload a photo of their listening environment.

Open-ended data are analysed through deductive thematic analysis, using the Common Sound Scenarios (CoSS) framework. Rating data are examined relative to listening intentions, and to each other.

Preliminary data will be presented, to understand everyday dynamic listening intentions of HA users, and identify the most common listening intentions and their importance. In capturing diverse, ecologically valid hearing experiences, findings will support future research investigating how behavioural and acoustic data might be used by HAs to predict listening intentions.

P13. Cross-modal representations in the auditory cortex of deaf and hearing individuals

Manini B (1, 2), Grin K (1), Eimer M (3), Evans S (4, 5), Tamè L (6), Woll B (1), Bertani R(1), Ercelik D (1), Core L (1), Gao Y (1), Longo M (3), Vinogradova V (1, 7), Cardin V (1)

1. Deafness Cognition and Language Research Centre and Department of Experimental Psychology, University College London, London, UK
2. School of Science, College of Science and Engineering, University of Derby, Derby, UK
3. Department of Psychological Sciences, Birkbeck, University of London, London, UK
4. Neuroimaging Department, Institute of Psychiatry, Psychology & Neuroscience, King's College London, London, UK
5. Institute of Cognitive Neuroscience, University College London, London, UK
6. School of Psychology, University of Kent, Kent, UK
7. Center for Language and Brain, HSE University, Moscow, Russia

Studying early deafness offers a unique insight on how sensory experience shapes brain organisation. In this work, we investigated whether the auditory cortex in congenitally deaf individuals can support higher-order cognitive functions when its typical sensory input is absent.

In congenitally deaf individuals, auditory regions respond to visual and somatosensory stimulation. This is known as cross-modal plasticity. Beyond these sensory responses, studies have shown that auditory areas are also recruited during cognitive tasks in the visual and tactile modality. This suggests that, in the absence of auditory input, these regions may take on roles that extend beyond sensory processing. Yet it remains unknown what role the auditory cortex plays during executive processes. One possibility is that activity in the auditory cortex during visual or tactile cognitive tasks may contribute to processing sensory features required to complete the task. Alternatively, the auditory cortex may represent higher-level information, such as attentional state or task rules.

To address this question, we conducted a cross-modal fMRI delay-to-match experiment in 13 deaf and 18 hearing participants, manipulating sensory modality (visual, tactile) and task-relevant information (temporal, spatial). We applied Representational Similarity Analysis to investigate whether the auditory cortex of deaf individuals represents sensory features or task-related information.

We found that the auditory cortex of deaf individuals represents tactile temporal features. We observed representations of sensory modality and task information in deaf and hearing participants. This overlap indicates that the reorganised auditory cortex in deaf individuals may draw on representational mechanisms already present in hearing brains, rather than developing entirely new ones. Results also point to an active role of the auditory cortex in cognitive processes that go beyond the maintenance of sensory representations.

P14. No evidence of a benefit for speech-in-noise intelligibility from recalibration to auditory-visual asynchrony

Vuong QC (1), Rees A (1), Mason C (1), Avery-Hughes I (1), Sumner C (2), Knight S (1)

1. Newcastle University, Newcastle, UK
2. Nottingham Trent University, Nottingham, UK

Understanding speech in noise is important for social communication. In these situations, people use visual information synchronised to the speech signal, such as lip movements, for speech comprehension. The extent to which auditory and visual information contribute to comprehension depends on the synchrony between the two signals. We previously demonstrated that participants recalibrated their percept of synchrony over ~30s after a visual-leading asynchrony was introduced into continuous audiovisual (AV) speech (i.e., the asynchronous speech was perceived to be more synchronous over time). Here we ask whether this recalibration improves speech-in-noise intelligibility. On each trial, participants (N=32) were presented with an AV recording of a talker on a monitor and over headphones. The talker spoke a series of IEEE sentences, with a short pause (~1s) between sentences. We manipulated AV synchrony (synchronous, asynchronous) and stimulus duration (short, long). Synchronous stimuli remained synchronous throughout. For asynchronous stimuli, a 240ms visual-leading asynchrony was introduced after two synchronous sentences (~5-7s). Short stimuli consisted of 4-5 sentences (~20s) and long stimuli consisted of 8-9 sentences (~40s). Stimuli were presented with a competing auditory stream of non-matching IEEE sentences and embedded in noise based on the long-term average spectrum of all sentences. SNR was individually determined for 50% accuracy. Participants repeated the final sentence and were scored on number of correct keywords. Responses were significantly less accurate in the asynchronous (40%) vs. the synchronous (56%) condition. This asynchrony effect was the same irrespective of stimulus duration, and accuracy did not differ across durations. Thus, while asynchrony reduces intelligibility, the similar magnitude of the asynchrony effect for both durations suggests that the recalibration observed in the detection of synchrony per se does not improve intelligibility.

P15. Does voice familiarisation training improve speech-in-speech intelligibility for older adults with presbycusis?

Patel B, Smith HJ, Rosen S, Holmes E

University College London, London, UK

Difficulty following conversations in noisy environments is a widespread challenge, particularly for those with presbycusis (i.e., age-related hearing loss). One factor that can facilitate speech intelligibility in such environments is voice familiarity, derived from naturally familiar or laboratory-trained voices. Voice familiarity has been shown to benefit both younger and older adults with normal hearing. In the present study, we assessed whether adults with presbycusis benefit from voice familiarity to a similar extent as adults with normal hearing. We recruited native English speakers over the age of 55, with no cognitive or neurological impairments. The control group had clinically normal hearing. The test group had mild-to-moderate sensorineural hearing loss and did not use hearing aids. Participants were first familiarised with three male voices. They then completed a training phase in which they identified these three voices and received feedback. Following training, participants completed an intelligibility test, identifying words in a target sentence in the presence of a competing sentence. The target voice was either one of two trained (familiar) voices or a novel voice. A total of six male voices were available, which were counterbalanced across participants, so that each voice was presented as familiar to some participants and novel to others. We measured speech reception thresholds (SRTs) by varying the target-to-masker ratio to estimate the 50% threshold and compared SRTs for familiar and novel target voices. Consistent with previous research, our preliminary results indicate a familiar-voice benefit to SRTs in older adults with normal hearing. When data collection is complete, we will test for differences in the familiar-voice benefit between groups. The results will reveal whether hearing loss affects how voice identity information is used during speech perception, with practical implications for improving communication for people with presbycusis.

P16. Towards modelling the impact of spread of excitation on classical psychophysical paradigms with cochlear implant users

Chatiza R, Shahidi L, Carlyon RP, Guérit F

Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK

Cochlear implant (CI) users typically understand speech in quiet conditions, but in noisy conditions even the most successful struggle. One contributing factor is channel interactions, whereby the electrical fields generated by multiple CI stimulating electrodes overlap, leading to unintended co-stimulation of auditory nerve fibres. Auditory-nerve computational models could help us design and evaluate stimulation methods that reduce channel interactions, allowing rapid testing of varied stimuli. However, these models are inspired from animal data and are rarely validated against CI psychophysical literature. Here, we validate a computational model by comparing it to several published CI psychophysical datasets.

We select papers from the CI psychophysical literature with within-participant manipulations (e.g. the effect of temporal and spatial separation between two pulse trains on loudness summation) that investigated interactions both within and across channels. We then simulate them via the model by Brochier et al. (2022, IEEE TBME 69(11)) and test different metrics to compare the data against the modelled neural activation patterns. Finally, we investigate the impact of changing the model selectivity and time constants (refractoriness, central integration) on the fit between modelled neural activity and psychophysical data.

Ongoing results show modelled outputs that are overall in range of the data found in the literature, as well as expected and unexpected differences between the modelled and published within-participant effects. This suggests necessary improvements in the model parameters and/or improvements in the metrics used to map modelled neural activity to psychophysical data.

By improving the comparison between the output of CI computational models and known within-participant effects, computational models can improve our understanding of the complex impact of channel interactions on speech perception with CIs.

Brochier et al., IEEE Trans Biomed Eng, 2022. DOI: 10.1109/TBME.2022.3167113

P17. Head-centred sound localisation behaviour reveals head and world centered reference frames in auditory cortex

Aghili N, Town S, Bizley J

UCL Ear Institute, University College London, London, UK

Previous studies in passively listening animals have suggested that auditory cortical neurons represent sounds in both head-centred and world-centred spatial reference frames. We developed a sound localisation task that dissociates head-centred and world-centred coordinates and used this to examine spatial tuning in auditory cortex during head-centered localisation behaviour.

Three ferrets were trained to initiate trials from a central platform surrounded by 12 speakers. For two animals, target sounds were presented from the front (0°) and back (180°) of the head, while for the third animal sounds were presented from the left (-90°) and right (+90°). Following training, the orientation of both the central platform and target speakers was rotated between sessions, preserving the behavioural contingency in head-centred coordinates requiring animals ignored world-centred spatial locations. Probe sounds were presented from the remaining speakers to characterise spatial tuning across platform orientations.

Neural activity was recorded from auditory cortex using chronically implanted 32-channel electrode arrays. Spikes were sorted using MountainSort4 and synchronised with behavioural and stimulus events. Across 3517 putative units, 702 were stimulus responsive, and 667 were spatially tuned. Of these, 585 units were head-centred and 82 were world-centred.

Spatial tuning across the population was also biased towards locations associated with the animals' training experience, with preferred locations corresponding to positions behind the animal relative to each animal's trained platform orientation.

We observed that world-centered representations may emerge through training: when we compared neural responses at the trained platform angle, and at 180° rotation (i.e. identical target speakers but opposite behavioural contingencies) many otherwise head centered neurons preserved the world-centered tuning for this location.

P18. Language Proficiency Drives Listening-Related Fatigue in Those Who Communicate in Their Second Language

McGarrigle R (1), Havelka J (1), deBruin A (2)

1. University of Leeds, Leeds, UK
2. University of York, York, UK

Individuals who listen in their second language ('L2 listeners') are disproportionately impacted by adverse acoustic conditions; a phenomenon coined the 'non-native speech-in-noise disadvantage'. Evidence also suggests that L2 listening taxes cognitive resources to a greater extent than in L1 listeners. Very little, however, is known about the longer-term consequences of this sustained cognitive engagement. Listening-related fatigue refers to the daily life subjective experience of tiredness and exhaustion from effortful listening.

The current study recruited a large sample of L2-English listeners (N = 256) to explore linguistic and demographic factors that predict listening-related fatigue. Approximately half (n = 126) of this sample reported an Indo-European (IE) language (e.g., Italian) as their first language, whereas the other half (n = 130) reported a non-Indo-European (non-IE) language (e.g., Vietnamese) as their first language.

Non-IE L2 listeners reported higher overall listening-related fatigue than both IE L2 listeners and monolingual controls. Both self-rated and objective English language proficiency were found to be significant predictors of listening-related fatigue in L2 listeners. Additionally, language proficiency moderated the effect of L2 use on listening-related fatigue; individuals who reported increased L2 use were more likely to experience increased fatigue, but only if their language proficiency (both self-rated and objective) was low and/or moderate. Finally, exploratory mediation analysis suggested that the effect of L2 listener group (IE vs non-IE) on listening-related fatigue was largely attributable to lower language proficiency in the non-IE L2 group.

To conclude, language proficiency appears to be a strong determinant of listening-related fatigue in L2-English listeners. When L2-English proficiency is low, increased L2-English usage is associated with heightened listening-related fatigue.

P19. The social and temporal influences on attention in audiovisual scenes

Griffiths ML, Beesley T, Stewart H

Lancaster University, Lancaster, UK

There is more to listening than what an individual hears. Visual speech describes the visual information we receive in communication, such as through emotions and lip movements. It plays an important role in facilitating dialogue, particularly in more challenging and noisy environments. The N2ac and N2pc are ERPs that are thought to reflect focusing of auditory and visual information in selective attention towards lateralised targets, respectively. However, the extent to which these ERPs are modulated by social and temporal features of visual information together is mostly unknown, despite the benefit of audiovisual information on speech perception being well established.

This study aims to uncover more about the role of selective attention and our understanding of why visual information is so beneficial in listening scenarios. We recruited 32 neurotypical hearing young adults. Participants completed hearing, listening, speechreading, and vision assessments. Participants were asked to locate an auditory target numeral (left, right, or absent) using a key press while videos played simultaneously to the left and right of fixation. These videos were either: scrambled faces, still faces, faces with audiovisual incongruent lip movements, or faces with audiovisual congruent lip movements. This allowed us to create lateralised targets to observe the N2pc/N2ac using EEG. Gaze was recorded using eye-tracking, allowing us to remove trials without lateralised visual stimuli.

Preliminary EEG results (n = 25) suggest the N2ac is only present in the scrambled and audiovisual incongruent conditions. We observed larger N2pc peaks for passive and audiovisual congruent conditions, suggesting participants attended more to the visual information here. Social and temporal comparisons are not currently significant. ~18% trials have been flagged for removal due to the gaze data (n = 4). We will present the impact of this on the full neural data and explore predictors of these changes.

P20. Perceptual distinctiveness of place and voicing mispronunciations in hearing primary school aged children

Kaduk K (1), Holt R (2), Blair L (1), Howard L (1), Stewart HJ (1)

1. Lancaster University, Lancaster, UK
2. Macquarie University, Sydney, Australia

Not all speech errors are equally obvious. Some studies suggest that changes in place of articulation (i.e., where in the vocal tract a consonant is produced) are particularly difficult to detect, while others indicate that voicing changes (i.e., whether the vocal cords vibrate) may be especially challenging. This work presents part one of a two-stage research project investigating how children perceive subtle speech errors in continuous narratives.To inform development of our narratives, we first assessed the perceptual distinctiveness of mispronounciations differing in place of articulation or voicing. Initial consonants of high-frequency nouns (e.g., ‘bag’) were manipulated by changing either place of articulation (‘dag’) or voicing (‘pag’). We used a controlled three-alternative forced-choice (3AFC) behavioural paradigm and measured children’s reaction times (RT) and accuracy (ACC) in identifying the “odd-one-out”. We hypothesised that children detect the deviant words at above chance levels and their perceptual sensitivity to mispronunciations would differ for place and voicing contrasts. We also explored the relationship between performance and age, vocabulary, and processing speed. We present preliminary data from twenty-two typically developing, hearing children aged 6-11 years. The current data suggests that children detected the mispronounced word, with accuracy significantly above chance. Currently there is no significant difference in ACC between place of articulation and voicing manipulations, but reaction times for place of articulation manipulations were significantly faster when compared to voicing. Exploratory analyses show that age significantly predicts both RT and ACC, with RT decreasing and ACC increasing with age. These preliminary findings have informed the design of our subsequent Event-Related Potential (ERP) study, in which we are using ERPs, particularly the N400 component, to examine how children process ambiguous speech in real time.

P21. Within- and between-session variability in DPOAE measurements in an individual-differences study of auditory perception

Winge N, Chousidis C, Mason R

University of Surrey, Guildford, UK

Distortion product otoacoustic emissions (DPOAEs) provide an objective index of cochlear function, but their interpretation in individual-differences research depends on measurement stability. In this study, DPOAEs were collected as part of a wider experiment examining relationships between cochlear function and auditory perception, including intensity and frequency discrimination thresholds measured at matched frequencies. The present analysis focuses on within-session repeatability and between-session variability in the DPOAE measurements.
Forty participants completed two test sessions on the same day. In each session, three DPOAE measurements were obtained from each ear across 11 frequencies, with a pass criterion of signal-to-noise ratio of at least 6 dB. Pure-tone audiometric thresholds were also measured to characterise hearing sensitivity. For each participant, ear, frequency, and session, repeated measurements were summarised using the median valid DPOAE level. Within-session variability was estimated from the spread of repeated measurements within each session, while between-session variability was estimated from the difference between Session 2 and Session 1 median DPOAE levels.
Preliminary analyses suggest little evidence of a systematic mean shift between sessions, indicating that DPOAE levels were not consistently higher or lower at the second session. However, individual session-to-session differences appeared larger than repeat variability within a session.
Ongoing analyses will examine how variability differs across frequency and ear, and whether session-to-session DPOAE changes relate to session interval, audiometric threshold, or behavioural measures of auditory perception. These findings will help clarify how DPOAE measurements should be summarised and interpreted when relating cochlear measures to perceptual performance.

P22. Does optogenetics based inactivation of the secondary auditory cortex influence vowel discrimination in noise in ferrets?

Hallou C, Madanat L, Watt F, Bizley J

UCL Ear Institute, University College London, London, UK

The ability to focus on specific sounds in complex noisy environments is critical for survival and communication. Previous work has shown that inactivation of the middle ectosylvian gyrus (MEG), which contains the primary auditory cortex (AC), impairs vowel discrimination in noise (Town et al. 2023). Work with human subjects has shown that unlike primary AC areas, secondary areas in the AC successfully retain sound recognition capabilities in naturalistic background noise conditions (Kell et al. 2019). Here, we asked whether the posterior ectosylvian gyrus (PEG), a tonotopically organised field of secondary auditory cortex, contributes to vowel discrimination in noise, and whether this differed between white noise and more naturalistic background noises.

In this study, ferrets were trained to discriminate the vowels /u/ and /e/ by responding at one of two lateral spouts in a two-alternative forced choice task. Vowels were presented at different attenuations, in silence, or in a noise background. Noises were either white noise, or natural sounds drawn from a database (collated by Mishra et al 2021). Ferrets were bilaterally injected with a viral vector (AAV-mDlx-ChR2_mCherry) in the PEG and implanted with a custom-made 3D printed light guide to allow light driven inhibition of auditory cortex by stimulating inhibitory interneurons.

We are currently investigating whether silencing PEG affects listening performance by introducing laser stimulation on 50% of trials, across different signal-to-noise ratio values (ranging between -5 to 15dB). We hypothesise this stimulation will have an effect on vowel discrimination in noisy listening conditions, particularly in more complex naturalistic background noises, while having no effect on performance in silence. Data collection is ongoing, although preliminary analysis in the first animal suggests discrimination in naturalistic noises is impaired.

Town et al., J Neurosci, 2023. DOI: 10.1523/JNEUROSCI.1426-22.2022
Kell et al., Nat Commun, 2019. DOI: 10.1038/s41467-019-11710-y
Mishra et al., PLoS One, 2021. DOI: 10.1371/journal.pone.0238960

P23. The first Cadenza lyric intelligibility prediction challenge

Graetzer S (1), Akeroyd MA (2), Bannister S (3,4), Barker J (5), Cox TJ (1), Roa Dabike G (5), Fazenda B (1), Firth J (2), Greasley A (3), Vos RR (1), Whitmer WM (2)

1. Acoustics Research Centre, University of Salford, Manchester, UK
2. Hearing Sciences, School of Medicine, University of Nottingham, Nottingham, UK
3. School of Music, University of Leeds, Leeds, UK
4. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
5. Department of Computer Science, University of Sheffield, Sheffield, UK

Listening to music has health and well-being benefits, and understanding lyrics is part of music enjoyment. However, hearing loss and hearing aids can reduce listeners' ability to understand lyrics. This is a significant concern; more than 430 million people worldwide have disabling hearing loss. Improving and predicting lyric intelligibility is not trivial, and can be affected by vocal style, song genre, mixing production and listener hearing ability. The performance of speech intelligibility prediction metrics for sung lyrics is unreliable.

In the Cadenza Lyric Intelligibility Prediction challenge (CLIP1), entrants were asked to improve on baseline prediction models for different levels of hearing loss. The task was to predict lyric intelligibility from musical excerpts of 5-10 words. A dataset of 11,072 listener-labelled samples was provided, with two baseline systems; a use of Short-Term Objective Intelligibility (STOI) with an audio source separation model (HDemucs), and a Whisper-based Automatic Speech Recognition (ASR) system. Twenty-two teams submitted 27 systems, of which 16 outperformed the best (Whisper-based) baseline. Twenty-four systems used foundation models for high-level acoustic representation (including Whisper). Four systems used text encoders including Hugging Face T5 models. Some systems made use of features such as mel-frequency cepstral coefficients (MFCC), signal-to-noise ratio (SNR) and pitch. Four systems used data augmentation methods.

Analysis of the results indicated opportunities for improvement. Sources of variability in the data included singing style and accompaniment, which could be better modelled by larger, more varied datasets. Accordingly, the datasets for the second Cadenza Lyric Intelligibility Prediction challenge (CLIP2) feature thousands of excerpts from AI-Generated songs, simulating different listening scenarios and genres. CLIP2 will launch in June 2026. Submission deadline: 30th October 2026.

P24. Exploring noise sensitivity in autism through cortical speech tracking

Simon A (1), Linden J (1,2)

1. UCL Ear Institute, University College London, London, UK
2. Department of Neuroscience, Physiology and Pharmacology, University College London, London, UK

Many autistic individuals report significant difficulties understanding speech in everyday noisy environments, particularly in situations involving multiple competing sound sources. These auditory challenges may contribute to sensory overload, reduced social participation, and decreased quality of life. The neural signature underlying these listening difficulties remains poorly understood. One hypothesis is that this noise sensitivity reflects reduced suppression of background noise, leading to enhanced neural processing of task-irrelevant auditory input.

To investigate this hypothesis, we are comparing cortical speech tracking in autistic and non-autistic normal-hearing adults during a continuous speech listening task. Participants listen to a target speaker presented either in quiet or alongside a competing talker that they are instructed to ignore. Continuous EEG is recorded during listening, and Temporal Response Functions (TRFs) are computed to quantify cortical tracking of both target and distractor speech.

Data collection is currently ongoing. We hypothesise that cortical tracking of attended speech will be comparable across groups. In contrast, we expect autistic participants to exhibit enhanced cortical tracking of distractor speech relative to non-autistic participants. Such a pattern would suggest reduced suppression of irrelevant background speech and may provide a neural signature of noise sensitivity in complex acoustic environments.

In addition, we aim to examine whether neural measures of distractor-speech tracking are associated with self-reported autistic traits, sensory hyper- and hyposensitivity, and listening difficulties in daily life.

Overall, this work seeks to advance our understanding of the neural mechanisms underlying auditory processing differences in autism and their relationship to real-world listening experiences.

P25. Perceiving the absent: Auditory expectations generate illusory percepts under inattention

Verbickas A, Clarke J

University of West London, London, UK

Perception reflects an interplay between sensory evidence, prior expectation, and attention. Predictive-processing accounts propose that perception depends on precision-weighted integration of top-down predictions and bottom-up sensory evidence, with attention regulating the relative influence of prediction errors and priors. Previous visual work suggests that strong expectations can generate illusory perceptual reports when an expected stimulus is absent, including letters, colour patches, and faces under inattention; this phenomenon has been termed expectation awareness (Clarke, 2025; see also Aru et al., 2017, 2018; Erol et al., 2018; Mack et al., 2016). Here, we tested whether auditory expectations can similarly induce illusory perceptual experiences and whether this effect depends on attention. Twenty participants completed a dichotic listening task in which a 200ms stimulus was repeatedly presented to one ear while participants performed a pitch-discrimination task in the other. Attention was manipulated across inattention, divided, and full-attention conditions. On critical trials, the expected sound was omitted. Participants reported whether they perceived a sound and rated its clarity using the Perceptual Awareness Scale (PAS). False-positive reports differed significantly across attention conditions, Cochran's $Q = 16.71$, $p < .001$, 65% inattention, 20% divided and 5% full attention. Planned McNemar comparisons showed higher false-positive rates in inattention than in divided attention ($p = .012$) and full attention ($p < .001$). PAS ratings showed a similar pattern, Friedman $\chi^2(2) = 17.63$, $p < .001$. These findings show that auditory expectations can generate illusory perceptual experiences without sensory input, particularly under inattention. The results extend expectation-awareness effects into the auditory domain and are consistent with predictive-processing accounts in which attention modulates the balance between prior expectations and sensory evidence.

Clarke, Constructing Experience: Expectation and Attention in Perception, 2025. DOI: 10.1017/9781009588577
Aru & Bachmann, Conscious Cogn, 2017. DOI: 10.1016/j.concog.2017.06.017
Aru et al., Conscious Cogn, 2018. DOI: 10.1016/j.concog.2018.09.001
Erol et al., J Vis, 2018. DOI: 10.1167/18.10.1115
Mack et al., Conscious Cogn, 2016. DOI: 10.1016/j.concog.2015.12.006

P26. Association between the Envelope Following Response (EFR), loudness scaling, and speech perception

Vinceslas L (1), Verhey JL (2), Drga V (1), Yasin I (1)

1. Department of Computer Science, University College London, London, UK
2. Department of Experimental Audiology, Otto von Guericke University Magdeburg, Magdeburg, Germany

Hyperacusis describes the condition where, despite normal auditory thresholds, mid-level sounds are perceived as uncomfortably loud. It is suggested that a component of hyperacusis may be associated with cochlear synaptopathy. Some studies suggest that the envelope following response (EFR), an electroencephalography (EEG)-based measure, may be used as an indicator of cochlear synaptopathy (Biest et al., 2023). The current study investigated the correspondence between measures of loudness sensitivity and growth and the EFR in a young population. Participants undertook standard audiometry and tympanometry followed by an assessment of their loudness experience using the Hyperacusis Questionnaire (HQ). In addition, categorical Loudness Scaling (CLS) was used to obtain a measure of loudness growth functions from which an estimate of the rate of growth of loudness can be obtained for such a population. All participants had audiometric thresholds (0.25- 8kHz) within normal hearing limits (<=20 dB HL) and a range of loudness sensitivity results when assessed with the HQ. The EFRs were measured in response to 300-ms rectangular-amplitude modulated stimuli (carrier of 2347 Hz). Results are discussed with reference to speech perception, loudness perception, and inferred cochlear synaptopathy (using the EFR measure).

P27. Comparison of eCAP estimates of neural health in cochlear implant listeners

Haywood, NR

University of Cambridge, Cambridge, UK

This planned research will compare four eCAP measures – panoramic eCAPs (PECAPS, Garcia et al., 2021), inter-phase gap (IPG) effects (e.g., Ramekers et al., 2014), polarity effects (e.g., Undurraga et al., 2010), and the Failure Index (Konerding et al., 2025). Previous research has suggested each measure may be informative of peripheral neural survival / health in cochlear implant recipients. In the first instance, we aim to test 12 listeners with Cochlear devices, and compare these measures both across electrodes within individuals (e.g., whether all measures return similar estimates for regions of poor neural health, such as 'dead regions') and between individuals (e.g., whether eCAP estimates of neural health correlate with behavioural outcome measures of speech in noise and spectro-temporal processing). We will also compare the four proposed eCAP measures to more conventional clinical eCAP measures – such as eCAP threshold.

eCAP measurements will be recorded via clinical software (Custom Sound 6.0). IPG effects will be measured from eCAP amplitude growth functions (AGFs) for pulses with either a 7 or a 58 μs IPG. Polarity effects will be measured from AGFs for cathodic- and anodic-leading biphasic pulses (58 μs IPG). The Failure Index (the ratio between stimulating current/charge and eCAP amplitude) will be estimated from the AGFs obtained for IPG and polarity effect measurements.

We will present details of the final test battery for this study, and the preliminary data obtained. The ultimate goal of this project is to develop a test battery to estimate peripheral neural health, which could be used to assess outcomes from advanced therapies (gene therapies, stem cell therapies).

Garcia et al., J Assoc Res Otolaryngol, 2021. DOI: 10.1007/s10162-021-00795-2

Ramekers et al., J Assoc Res Otolaryngol, 2014. DOI: 10.1007/s10162-013-0440-x

Undurraga et al., Hear Res, 2010. DOI: 10.1016/j.heares.2010.06.017

Konerding et al., J Neurosci, 2025. DOI: 10.1523/JNEUROSCI.0954-24.2024

P28. Towards better cortical auditory evoked potential biomarkers for schizophrenia risk: investigations in a mouse model

Lax G (1), Lu C (1), Bramon E (2), Schmack K (2,3), Linden JF (1,4)

1. UCL Ear Institute, University College London, London, UK
2. Division of Psychiatry, University College London, London, UK
3. Francis Crick Institute, London, UK
4. Department of Neuroscience, Physiology & Pharmacology, University College London, London, UK

Cortical auditory evoked potential (CAEP) measures (e.g., P50/N100 suppression, auditory steady-state responses, P300 amplitude, mismatch negativity) differ significantly between schizophrenia patients and controls (Wang et al. 2022 Translational Psychiatry). These CAEP measures have potential as biomarkers but high inter- and intra-individual variability limits their clinical use. Improved accuracy and robustness to variability could make CAEP measures useful for early diagnosis of schizophrenia.

Here, we present a preclinical proof-of-principle showing that mice with a genetic risk factor for schizophrenia can be differentiated from controls using a novel combinatorial CAEP biomarker. CAEPs were recorded in anaesthetised mice during presentations of tone pip sequences varying in sound level and inter-tone interval. We obtained robust measures of CAEP gain and adaptation by quantifying level-dependent and interval-dependent growth in P1-N1 and N1-P2 amplitude. Previously we reported abnormalities in all four of these measures in the Df1/+ mouse model of genetic risk for schizophrenia (Lu & Linden 2025 Translational Psychiatry). We reanalysed this data (n=40 WT, n=12 Df1/+) using leave-one-out cross-validated linear discriminant analysis to find the optimal combination of measures that maximised differences between groups. The combinatorial model achieved 82.7% correct classification (false positives/negatives, 15%/25%), performing better than the best of the individual AEP measures (interval-dependent P1-N1 growth, 75% correct classification, false positives/negatives 22.5%/33.3%). The combinatorial model also yielded a notably higher effect size for separation of animal groups (Cohen's d=1.75 vs d=1.19).

In ongoing extensions of this work, we are measuring CAEPs in awake animals and investigating correlation of CAEP measures with behavioural measures of hallucination-like percepts in mice (Schmack et al. 2021 Science).

Wang et al., Transl Psychiatry, 2022. DOI: 10.1038/s41398-022-01860-x
Lu & Linden, Transl Psychiatry, 2025. DOI: 10.1038/s41398-024-03218-x
Schmack et al., Science, 2021. DOI: 10.1126/science.abf4740

P29. Current focussing with cochlear implants: on efficiently assessing and optimizing its impact

Guérit F, Garcia C, Deeks JM, Carlyon RP

Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK

Cochlear-Implant (CI) users struggle to understand speech under noisy conditions. This has long been attributed to channel interactions: a given neuron not only responds to electrical stimulation from the closest electrode but is also sensitive to stimulation from electrodes located further away. Attempts to sharpen neural excitation patterns (for example, replacing the clinical monopolar stimulation mode with tripolar and phased-array methods) inject current via one intra-cochlear electrode and return it via opposite-polarity stimulation of other electrodes. These approaches have met mixed success, partly because of the lack of accurate outcome measures.

To tackle this, we developed a psychophysical technique that measures both the magnitude and polarity of stimulation along the neural excitation pattern. It involves adding a subthreshold localized probe to an ongoing baseline stimulus and testing whether CI users can detect that pedestal change. However, it is limited by how quickly one can obtain measurements, precluding any “listener-in-the-loop” optimization, and reflects a change detected at a central level rather than at the auditory nerve (AN). We therefore investigate both a faster implementation and an AN electrophysiological equivalent (using electrically evoked compound action potentials, eCAPs).

For the faster psychophysical implementation, we investigate a Bayesian active learning approach that allows more efficient sampling of the parameter space of baseline-probe relative distances. For the eCAP approach, we investigate how best to remove CI artefacts and how to handle variability in overall eCAP responses across CI users and stimulation electrodes. Pilot results indicate that both methods can reduce testing time compared to our standard method. We are currently assessing their impact on the precision of the measured excitation patterns.

P30. A web-based listening test system and its validation for remote testing and clinical practice with cochlear implant recipients

Goehring T (1,2), Carlyon RP (1)

1. Cambridge Hearing Group, MRC Cognition and Brain Sciences Unit, University of Cambridge, Cambridge, UK
2. Department of Otorhinolaryngology, Head and Neck Surgery, University Hospital Zurich, University of Zurich, Zurich, Switzerland

Web applications are increasingly used in research and clinical audiology, driven by the development of mobile, remote technology and strong demand. Remote applications have potential to increase statistical power, accessibility and diversity in research and to support clinical audiology services. We aimed to bring these advantages to cochlear implant (CI) research and clinical practice and developed a web-based listening test system called AUDITO. Building on wireless streaming technology and smart devices, AUDITO can be used to flexibly implement and administer a wide range of listening tests remotely or in-the-lab. The system was designed to be easy to use without software programming. Experimenters can exchange listening test setups and stimuli sets within the system to facilitate collaboration. Technical checks were implemented to ensure signal quality over wireless streaming. A first validation study for CI research was performed with 20 experienced CI listeners. Comparisons of interest included the presentation of stimuli via cable vs Bluetooth streaming, and remotely vs in-the-lab. Three listening tests were implemented to measure speech perception, digits in noise and spectro-temporal resolution. A questionnaire was administered to collect user feedback. The system worked reliably with various setups including desktop and laptop computers, tablets and smartphones. Test results were consistent between listening modalities across conditions confirming the validity of AUDITO. User feedback was positive for system usability and function, while acoustic quality was not reported to be compromised via streaming. A clinical evaluation study is ongoing, with early results showing mixed results and some disagreement between in-clinic and at-home testing. The AUDITO system facilitates data collection, enables research collaboration, and has potential to support clinical audiology to improve accessibility, inclusion and save resources.

P31. Sex differences in forward-masking ABR Wave I recovery coincide with reproductive transitional stages in CBA/Ca Mice

Lam CC, Nolan LS

King's College London, London, UK

Sex as a biological variable (SABV) is increasingly recognised as a crucial factor in auditory physiology and susceptibility to age-related hearing loss. While conventional auditory brainstem response (ABR) metrics provide baseline characterisations of hearing sensitivity, they offer limited insight into short-term adaptation within the early auditory pathway. Here, we investigated whether SABV affects the temporal recovery dynamics of peripheral auditory processing across key reproductive transitional stages.

Using males and females CBA/Ca mice, forward-masking (FM) ABR paradigms were conducted at three key female reproductive transitional stages: pre-puberty (postnatal day P21), young adulthood (P53), and middle-age (P270). ABR Wave I responses to probe clicks at high (50 dB SL) and low (20 dB SL) intensities were measured at variable masker-probe intervals (MPI) from 4 to 64 ms following a 50 ms broadband noise masker. Probe responses were normalised to a non-masked reference click at 150 ms post-masker offset for the comparison of the functional recovery time course.

When comparing unmasked reference amplitudes, female mice exhibited significantly higher Wave I amplitudes than males at high stimulus intensities starting in young adulthood, and at both high and low intensities by middle age. In the recovery time course of masked probe responses, significant sex difference emerged exclusively in young adult (P53) mice. Sexually mature females demonstrated a delayed Wave I amplitude recovery trajectory compared to males. This difference was primarily localised to the rapid recovery phase (<50 ms MPI) at low intensity and extended to the slow recovery phase (>50 ms MPI) at high intensity. These sex differences were absent before sexual maturation at pre-puberty (P21), and after reproductive decline at middle-age (P270). These findings demonstrate that gonadal hormones may play a role in the modulation of short-term adaptation within the peripheral auditory pathways.

P32. Assessing suitable fNIRS hyperscanning paradigms in a naturalistic classroom laboratory

Salazar G (1), O'Hanlon BL (1), Usherwood B (1), Howard L (1), Anderson C (1), Ainsworth S (2), Stewart HJ (1)

1. Lancaster University, Lancaster, UK
2. Manchester Metropolitan University, Manchester, UK

In the classroom, external behaviours alone cannot show whether children are selectively processing speech content at the cortical level ('listening'), or simply passively detecting sound at the sensory level ('hearing'). Enhanced neural synchronisation at temporal-parietal and temporal-frontal junctions has been shown to reflect this difference between listening and hearing. In this study, we are assessing this neural synchrony in different listening paradigms to assess the paradigms' suitability for naturalistic hyperscanning in classrooms.

We are hyperscanning dyads of one child (aged 6-11 years) and one adult; both hearing, neurotypical, and native English speakers. Here, we present our preliminary data from 24 dyads. Each participant wears 39-channel wireless fNIRS in a montage covering temporal-parietal and temporal-frontal junctions. We are using throat mics to capture the timing of speech from participants, to be entered as event triggers post hoc to the synchronised cortical data.

We are assessing three paradigms: In Paradigm 0 participants listen to a recording of speech stimuli together; Paradigm 1 involves static turn-taking, where the child speaks whilst the adult listens, and vice versa; Paradigm 2 involves dyadic conversing, where the child and adult converse freely. For each, we are calculating cortical shared listening within a dyad using correlation coefficient matrices for brain-to-brain (through wavelet transform coherence) and brain-to-speech synchronisation (through backwards modelling). In Paradigms 0 and 1 this is each listener's neural data. In Paradigm 2 this is the concatenation of when participants were listeners, using bootstrapping to eliminate potential bias introduced by concatenation seams.

We predict that child-adult dyads will show greater shared listening when congruent (true dyads) versus incongruent (randomised dyads). Paradigms supporting this prediction may be considered suitable for naturalistic hyper-scanning experiments.

P33. Design and validation of a synchronisation system for different sensors across time in a naturalistic classroom

O'Hanlon BL (1), Usherwood B (1), Howard L (1), Anderson C (1), Ainsworth S (2), Stewart HJ (1)

1. Lancaster University, Lancaster, UK
2. Manchester Metropolitan University, Manchester, UK

Paediatric auditory research can take place in real-world environments as classrooms (offering high ecological validity), or in lab-based environments (offering high precision). We have created a naturalistic classroom laboratory following the L3 Framework (Mealings, et al., 2023), providing a hybrid between ecological validity and precision for evaluating listening in dyadic (two-person) and tetradic (four-person) setups. However, it is difficult to use complex equipment in naturalistic spaces, such as neuroimaging and body motion-tracking, as it requires wireless synchronisation across different sensors. We have designed a synchronisation system that allows for simultaneous wireless eye- and motion-tracking, audio recording, and fNIRS hyperscanning between multiple participants, all without compromise to the naturalistic setting of our classroom.

To validate the accuracy and versatility of this synchronisation system using the above sensors simultaneously, we have conducted a methodological proof. Here, we use a 'finger'-tapping task with both a single and a tetrad participant setup in our classroom. Participants were asked to follow a visual digital metronome using sets of different coloured drums. There were three distinct playing conditions: 'Musical score' where participants played one drum; 'Four corners' where participants played multiple drums in an order dictated by visual cues; and 'Twister' where participants played multiple drums in an order dictated by auditory cues. In the tetrad setup, some participants were given a different metronome to follow at set intervals, providing motor-relevant hyperscanning measures for when participants were in-sync versus out-of-sync.

All data from each sensor was interpreted in relation to the timings of each participant's 'finger'-tapping on the correct drum. Here, we discuss this validation of our synchronisation system, our naturalistic classroom design, and its implication for future naturalistic research.

Mealings et al., Front Educ, 2023. DOI: 10.3389/feduc.2023.1185167

P34. The challenge of quantifying audiovisual integration of speech

Sumner CJ (1), Stacey PC (1), Smith S (2), Roeser J (1), Baguley T (1)

1. Nottingham Trent University, Nottingham, UK
2. Eaton Peabody Lab, Harvard University, Cambridge, MA, USA

Speech comprehension is often enhanced by observing a speaker's face. The size of the benefit of seeing the talker (over just hearing them) has been shown to differ across individuals and tasks. A logical scientific question is whether an observed difference in this benefit across experimental groups is attributable differences in unisensory processing, or indicates that there is some underlying change in the way information is integrated across the senses. However, distinguishing these explanations is challenging because the benefit is dependent on exact stimulus conditions and is highly non-linear.

Here we test the robustness of standard statistical inference methods for detecting changes in audiovisual integration. We use signal-detection theory-based models of audiovisual integration as a surrogate for real data. The models capture the observed non-linearities and allow us to vary the integration functions and thus provide a "ground truth" against which to evaluate the statistical tests. We applied a range of standard statistical tests (including linear models, generalised linear models, and linear models of speech reception thresholds) to paired groups of simulated data, where groups can differ in unisensory processing and the underlying integration function.

Standard statistical tests frequently reported significant differences in visual benefit across groups of simulated data when there were no differences in the underlying integration functions. Rather, significant differences in visual benefit often arose from group differences in unisensory speech intelligibility. This suggests that these tests were not well suited to detecting changes in integration.

In summary, significant differences in the benefit of seeing a talker arising from conventional statistical inference must be interpreted with caution: the difference in benefit may be real, but it may not reflect any change in the underlying cue integration.

P35. Physiological markers of auditory situational awareness in complex spatialised scenes

Sztandera JT (1,2), Poole KC (2), Shiell M (3), Picinali L (2), Chait M (1)

1. UCL Ear Institute, University College London, London, UK
2. Dyson School of Design Engineering, Imperial College London, London, UK
3. Eriksholm Research Centre, Snekkersten, Denmark

Detecting changes in acoustic environments is essential for situational awareness. It remains unclear whether the spatial location of such changes modulates automatic orienting and arousal mechanisms. We measured pupil dilation, pupil dilation rate, and microsaccade rate while listeners (n=25) heard complex, spatialized auditory scenes rendered over headphones using individualized HRTFs. Participants were naïve to the critical manipulation: the appearance of a new source from one of five locations: front, left, right, back, or above. A subsequent localization task assessed perceptual spatial uncertainty.

Behaviorally-irrelevant source appearances elicited a cascade of ocular responses. Microsaccadic inhibition emerged from ~85ms after change onset, and was broadly comparable across locations, suggesting a location-invariant early orienting response to auditory change. Pupil dilation rate increased from ~200ms, followed by a phasic pupil dilation response from ~400ms, indicating engagement of arousal-related systems. Pupil responses were modulated by source location: changes from front/left/right elicited larger dilation than changes from above, with back responses showing a similar but weaker reduction. Behavioral localization revealed substantial confusion for front/back/above locations. However, this did not mirror the physiological data, as front sources elicited pupil responses comparable to lateral sources.

These findings demonstrate that complex auditory scene changes recruit oculomotor and autonomic systems even outside the focus of task relevance. They further suggest a dissociation between early, location-invariant attentional capture indexed by microsaccadic inhibition and later, location-sensitive arousal indexed by pupil dilation. Spatial biases in auditory situational awareness therefore appear to emerge after initial change detection, shaping arousal and behavioral performance rather than the earliest orienting response.

P36. Radian: Measuring the rate of change of speech-in-noise performance across one year

Firth J (1), Sumner C (2), Akeroyd MA (1)

1. University of Nottingham, Nottingham, UK
2. Nottingham Trent University, Nottingham, UK

Age-related hearing loss (ARHL) in a common sensory impairment that has been associated with cognitive decline and poorer quality of life (QoL) in older adults. It is known that hearing loss increases with age, nevertheless, there is substantial variation and less is known about the rate of change of ARHL at the individual level. Previous longitudinal research examining the rate of change of hearing loss is limited and our recent unpublished scoping review highlighted an average annual decline of 0.5 dB (0.5 kHz) and 1.5 dB (4-8 kHz). However, this does not account for age-related changes, notably there is a steady decline in hearing loss during middle adulthood (e.g., 40s, 50s, 60s). The current study is aiming to determine if hearing deterioration can be mapped via performance on the triple digits-in-noise (TDIN) task over the course of a year. A convenience sample of normal-hearing adults and individuals with hearing loss are completing the online TDIN about once a week, some of them for as long as a year. We will present here the results, initial analysis of the week-by-week variety and the long-term trend. The results will use an exponential fit to examine the underlying constancy of the rate of change of ARHL, the acceleration with age, and the proportional rate of change. This research was funded by a Medical Research Foundation (MRF) Launchpad grant.

P37. Measures of Subclinical Hearing Function: Long-Term Test-Retest Reliability and Relations between Change Outcomes

Guest H (1), Dewey R (2), Kluk K (1), Millman R (1), Munro K (1), Murray C (1), Prendergast G (1), Stephen R (1), Plack C (1)

1. The University of Manchester, Manchester, UK
2. University of Nottingham, Nottingham, UK

Recreational noise is a key risk factor for preventable hearing loss in young people, yet pure tone audiometry (PTA) may be insensitive to early noise-related damage. This three-year longitudinal study examined candidate physiological, psychoacoustic and speech-in-noise markers of recreational noise exposure, focusing on their reliability and on distinguishing cochlear hair cell changes from potential synaptopathic effects.

220 adolescents (16–17 years at baseline; tested Nov 2021–Dec 2022) were assessed, with 199 (90%) returning after three years (Nov 2024–Jan 2026). Analyses included 191 otologically healthy participants. Testing was unilateral except for spatial speech tasks. Measures assessed cochlear function (conventional PTA 0.25–8 kHz, extended high-frequency audiometry 10–16 kHz, and 2f1–f2 distortion product otoacoustic emissions), auditory neural function (auditory brainstem responses to filtered chirps and ipsilateral middle ear muscle reflexes), and speech perception (digits in noise, spatially separated babble, and competing spatial digit streams). Change scores were calculated as post–pre differences or ratios. Test–retest reliability across timepoints was quantified using intraclass correlation coefficients (ICCC for consistency; ICCA for absolute agreement), and Pearson correlations assessed relationships between change measures.

Reliability was moderate for PTA (ICCC = 0.61) and high for extended high-frequency audiometry (ICCC = 0.85) and DPOAEs (ICCC = 0.89). Neural measures showed moderate to good reliability (ABR ICCC = 0.74; MEMR ICCC = 0.71). Speech perception reliability was poor for digits in noise and spatial babble (ICCC = 0.18–0.24), but moderate for speech-in-speech (ICCC = 0.62). Change measures showed no significant correlations across modalities, except within cochlear mechanical measures from similar frequency regions.

P38. Relating illusory tone perception to tinnitus

Barker JLR (1), Hoare DJ (1,2), Sereda M (1,2), Sollini J (1)

1. Hearing Sciences, School of Medicine, University of Nottingham, Nottingham, UK
2. NIHR Nottingham Biomedical Research Centre, Nottingham, UK

The Zwicker tone (ZT) is a transient auditory illusion induced at the offset of spectrally notched noise. Perception is not universal; however, a single 2007 study reported a significantly higher prevalence in people with tinnitus. No further research linking ZTs and tinnitus has been carried out, and this finding has not been replicated. Furthermore, whether any mechanistic association might be physiological or cognitive can only be speculated.

We measured prevalence of ZT perception in people with and without tinnitus, while exploring bottom-up and top-down factors that might explain this relationship. Bottom-up approach: We mapped the range of notched noise parameters that induce ZT perception in people with normal hearing and tinnitus. Top-down approach: We investigated the influence of expectation on ZT perception by comparing participant responses in ZT-naïve and primed conditions.

We report that the ZT is induced in an organised fashion over a wide range of parameters. In turn, we can compare incidence rates between tinnitus and non-tinnitus groups and explore variations in the successfully-inducing stimuli. Given the association of tinnitus with cognitive factors, we also report how priming affects ZT perception. Consequently, our results pertain to both bottom-up and top-down influences on illusory sound perception.

If ZT perception is correlated with tinnitus, it may present a powerful experimental model. While the mechanism through which the ZT arises is not understood, it stands to reason that it may be related to tinnitus production. As such, interrogation of the ZT could shine light on the neural origins of tinnitus. Furthermore, the induction of illusory percepts in normal-hearing subjects may prove a valuable psychophysical tool for tinnitus research.

P39. Pitch-based stream segregation of complex sounds in ferrets

Tarka VM (1,2), Diuba A (1), Hesketh-Bream M (1), King AJ (1), Walker KMM (1)

1. University of Oxford, Oxford, UK
2. University College London, London, UK

Pitch, the tonal quality of sound, is among the strongest acoustic cues for selective listening for human listeners. Though many animals perceive pitch, the role of pitch in auditory attention in non-human animals is largely unexplored. Auditory feature binding, a component of auditory attention, has often been investigated in humans using ABA- streaming paradigms in which temporally interleaved low (A) and high (B) pitched tones are presented to form a repeating ABA-sequence. When A and B are close in pitch, listeners perceive a single integrated auditory stream with a "galloping" rhythm, while large pitch differences produce a percept of two segregated A and B streams. Here, we adapted the classical ABA- streaming paradigm to create a novel behavioural task that captures pitch-based auditory streaming in ferrets. Three ferrets were trained in a go/no-go task to detect target oddballs and ignore distractor oddballs in a stream of alternating low (A) and high (B) pitched tones. The pitch of the B tones was held constant in a session, while the pitch of the A tones was either 3, 6, 11, or 14 semitones below the B tones on any given trial. If ferrets, like human listeners, segregate sound streams based on pitch cues, task performance should improve as the pitch separation between streams increases. We found that the animals could use the pitch of both pure tones and complex sounds to segregate competing streams. Overall performance was worse when pitch information was degraded, and the animals were better able to segregate streams formed of complex tones than pure tones. This is the first evidence that non-human animals use the pitch of complex sounds for selective listening.

P40. Exploring the contribution of spectrum and modulation in the EHF region to speech intelligibility

Stone MA, Prendergast G

Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK

Recent recordings of anechoic speech by Monson et al. (2012) and Hopkins et al. (2019) have indicated significantly higher spectrum levels in the Extended High Frequency (EHF) region, above 8 kHz, than previously. Perceptually audible contributions from this region, despite their low Sensation Level, introduce variability in speech intelligibility experiments (Polspoel et al., 2022). Additional sources of variability have historically been identified from gender and inter-speaker differences, making inter-test comparisons hard to make even with non-complex speech tests, such as digits.

Analysing both Monson's and the Hopkins' recording corpora and comparing to the Moore et al (2008) corpus, we identified a range of variability in both spectrum level and amplitude modulation (AM) depth in the EHF range, within and across corpora. In two experiments to identify digits spoken in a background noise, we simulated the range of these differences. Signal processing, manipulations were performed on the same target speaker and background noise throughout, thereby controlling for speaker variability. Additionally, we measured AM discrimination abilities in the EHF using the 'ELEVEN' test. This new test is performed at the low Sensation Levels of the target speech met in the intelligibility experiments. Additional properties of the ELEVEN signal are being narrow band, largely avoiding transducer response issues, as well as having very low AM in the 8-16 Hz rate range, an important contributor to speech intelligibility.

We cautiously report results showing that, aside from the already identified contribution of EHF spectrum differences, there is possibly a much smaller contribution of AM differences. Additionally, benefit from EHF speech energy shows a modest correlation with performance in the ELEVEN test, especially just above the 8-kHz EHF boundary. Caution is needed in the interpretation since we identify some procedural issues in the execution of the experiments.

Monson et al., J Acoust Soc Am, 2012. DOI: 10.1121/1.4742724
Hopkins et al., ARU speech corpus (University of Liverpool), 2019. DOI: 10.17638/datacat.liverpool.ac.uk/681
Polspoel et al., Ear Hear, 2008. DOI: 10.1097/AUD.0000000000001142
Moore et al., Ear Hear, 2008. DOI: 10.1097/AUD.0b013e31818246f6

P41. The effects of control and stimulus on the self-adjustment of frequency-dependent gain

Benecke J (1), Whitmer WM (1,2)

1. Hearing Sciences - Scottish Section, University of Nottingham, Glasgow, UK
2. School of Health & Wellbeing, University of Glasgow, Glasgow, UK

In conventional hearing-aid personalisation, clinicians cannot hear what their patients hear, and patients cannot often reliably detect or describe what they hear. Self-adjustment avoids this issue but requires user controls that adjust hearing-aid signal processing parameters to be effective, efficient and easy. In this study, we explored (a) the roles of interface complexity and stimulus type in the self-adjustment of hearing-aid gain, and (b) how well individuals can adjust one sound to match another to assess the same interfaces and stimuli. Adult hearing-aid users with mild to moderate symmetrical sensorineural hearing loss repeatedly adjusted the gain (a) to their preference from individual prescription (n = 41) and (b) to match their previous preferences from a random starting point (n = 32) using three interfaces representing different bass/mid/treble configurations and three stimuli (music, speech and speech-in-noise). The large interindividual variability in self-adjusted gains clustered into three patterns of deviation from initial prescription: increased relative bass, overall gain reduction, and close to initial prescription. There were no substantial effects of interface nor stimulus on self-adjustment reliability (median σ = 2.8 dB), whereas absolute sound-matching error increased with increasing interface complexity and centre frequency. Neither individual sound-matching accuracies nor behaviours predicted either self-adjusted gains or their reliabilities, highlighting differences between self-adjustment and its psychophysically equivalent task. Overall, these results show that many – but not all – hearing-aid users can adjust gains with reasonable reliability, and while it can be difficult to predict the behaviour from the individual, the individual applies a similar self-adjustment behaviour across different interfaces and stimuli.

P42. Towards ecologically valid measures of listening effort: a novel comprehension-based paradigm to assess cognitive load in noise

Bleeck S, Hallibone J

Institute of Sound and Vibration Research, University of Southampton, Southampton, UK

Conventional audiological assessments prioritize speech intelligibility, often failing to capture the underlying cognitive load required to comprehend speech in complex, real-world acoustic environments. To address this, our laboratory is developing frameworks to understand how hearing devices can utilize information about a user's cognitive state to minimize listening effort, independent of standard intelligibility enhancements.

We present a novel, ecologically valid listening comprehension paradigm designed to measure the cognitive impact of noise purely within the auditory domain. The methodology requires listeners to process 40-second spoken passages, followed by multiple-choice queries assessing both basic working memory recall and complex inference abilities. The focal task difficulty is individually calibrated to a set performance threshold, allowing for precise manipulation and measurement of cognitive load.

We are currently deploying this paradigm in a comprehensive study comprising up to 10 experimental conditions. This unified testing phase isolates and investigates four distinct auditory factors: The interaction between the focal cognitive load required for comprehension and the susceptibility to background speech interference. The specific disruptive impact of semantic content within background noise, comparing equivalent energetic maskers. The efficacy of advanced consumer audio devices (e.g., Apple AirPods Pro 2) in mitigating listening effort during multi-talker babble. The cognitive disruption caused by uniform versus temporally varying STM noise compared to natural speech maskers.

Data collection is currently underway. This presentation will detail the paradigm's methodological architecture and report initial findings, evaluating the sensitivity of this approach in distinguishing sensory resolution deficits from higher-order attentional and cognitive limitations.

P43. Developing a paradigm to assess the impact of cognitive status on cochlear-implant audiovisual benefit

Salorio-Corbetto M (1,2), Koubeh E (1), Glennon J (1), Fernández Ledesma M (1,3), Haywood N (1), Vickers DA (1)

1. Cambridge Hearing Group, SOUND Lab, Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK
2. Emmeline Centre for Hearing Implants, Cambridge University Hospitals, Cambridge, UK
3. European University of Madrid, Madrid, Spain
   corresponding author: ms878@cam.ac.uk

Lipreading supports speech-in-noise performance, but individual audiovisual benefit for speech recognition varies, even when performance for visual-alone presentation is accounted for. This project investigates the effect of cognitive status on audiovisual speech perception.

Participants undergo hearing and vision screening, speech-reading, auditory psychophysics (streaming and temporal fine structure use), and cognitive tests (fluid intelligence and visual span). The speech task was optimised by piloting (n=3, typical hearing - TH) stimuli which combined visual (Gaussian blurring parameter 15 and 45) and auditory (SPIRAL vocoding with 4, 8, or 16 activated electrodes - eN) degradations and two sentence corpora (BKB or IEEE) in a sentence-recognition task.

Participants rated their accuracy, the level of difficulty, and the usefulness of visual cues. Performance rose with eN until plateauing, but ceiling effects occurred. Profit from visual cues varied with performance level. Accuracy ratings mimicked the performance functions. Difficulty ratings varied with visual degradation depending on performance level. Visual-cue usefulness was lowest for the clearest condition for BKB, but not for IEEE sentences. In a follow-up pilot (n=3) with eN= 2, 4, or 8, floor effects occurred, and performance varied less across corpora.

Speech materials were replaced by a video version of the Coordinated Response Measure with nouns test (vCCRMn, Alampounti et al. 2025) to eliminate floor/ceiling effects and ensure equal audiovisual difficulty across keywords. The vCCRMn separates time coherence from viseme cues and explores target coherence effects. The test was modified to include variable visual clarity. 35 participants with TH and 35 with cochlear implants will be tested.

Funding: Supported by the Medical Research Council grant MR/S002537/1, the NIHR Biomedical Research Centre Devices and Advanced Therapies Theme, and Trinity College.

P44. Exploring the link between hyperacusis, anxiety and depression in a non-clinical UK cohort

Prendergast G (1), Couth S (1), Shehabi A (2)

1. Manchester Centre for Audiology and Deafness, School of Health Sciences, The University of Manchester, Manchester, UK
2. Birzeit University, Birzeit, West Bank, Palestine

Previous work conducted in the Palestinian territories during ongoing conflict described a positive relation between stress, anxiety, depression, and how strongly a person scores on a scale of hyperacusis (Shehabi et al, 2026). The data presented here explore the same relation in a group of young adults in the UK, aged 18-25. Participants were required to have no clinical diagnosis of stress, anxiety and depression. Rather, we sought to determine if the natural variation of these traits in the population has any relation to the percept of hyperacusis, despite the UK not, at the time of data collection, being a region of active armed conflict. Data were collected online via the Prolific platform and at the time of writing ethical approval has been granted and data collection will occur in the coming months. Data will be presented at the ASM meeting in September. The study was also pre-registered on the Open Science Framework.

Shehabi et al., Int J Audiol, 2026. DOI: 10.1080/14992027.2025.2521744

P45. Exploring the context-dependent nature of conversational success

Beith A, Nicoras R, Whitmer WM, Naylor G

University of Nottingham, Glasgow, UK

Increased likelihood of successful communication is a key outcome benefit of prescribed hearing-aid (HA) use. In previous work, we identified seven Facets of Success (FoS) that describe different ways in which a conversation can be successful (e.g., "being able to listen easily", "being engaged and accepted"). However, the relative importance of these facets is likely to depend on contextual factors of the conversation.

To address this, we developed a set of 49 conversational contexts that may influence conversational success. Candidate contexts were generated through a literature review and a HA user focus group. The focus group used images of contrasting conversational scenarios to elicit contextual factors that participants felt were important.

In an ongoing concept mapping study, participants with and without a hearing loss are asked to map FoS to the contexts in which they are most important. The resulting data will be used to identify a parsimonious set of conversational contexts that preserves the underlying structure of the relationships between context and success.

Preliminary findings will be presented, using correspondence analysis to produce geometric visualisations of the relationships between conversational context and FoS. An initial reduced set of conversational contexts will be discussed with consideration of how they will inform the development of models that better account for the context-dependent nature of successful communication.

P46. Factors influencing 10-year post implantation language skills in children with cochlear implants

Vickers D (1), Sear T (2), Hare J (2)

1. University of Cambridge, Cambridge, UK
2. University College London Hospital, London, UK

For infants with cochlear implants (CIs) many factors that affect variability in speech and language abilities are known but the importance of these factors and the interaction with other influencing factors change over the years and can be modified by changes in healthcare practice and in society.

The goal was to determine how the paediatric CI recipient profile changed over a 25-year period and to understand how these changes were influenced by clinical practice and changes in society and to relate changes to long-term (10-year) language outcomes. We conducted a retrospective analysis for a subset (167) of children implanted between 1998 and 2023 who had reached the 10-year post-implant stage. Demographic factors of home language, onset of severe-to-profound deafness (congenital, progressive, or acquired), age at implantation, socioeconomic status, aetiology and pre-implant hearing levels were collected. Logistic regressions were conducted to understand factors predicting language outcomes.

Preliminary analyses showed a significant increase in the proportion of children from non-native English-speaking families (influenced by population migration), and children with progressive onset of deafness (influenced by candidacy criteria). Age at implantation significantly decreased (due to greater surgical confidence, increased awareness of early implantation and newborn hearing screening). There was a significant reduction in socio-economic status believed to be related to recession, austerity, and population migratory trends. Regression analyses suggested that onset of deafness, age at implantation, year of implantation, income deprivation and parental education were key predictors of long-term language abilities.

If this complex and dynamically changing landscape of influential factors is well understood, appropriate interventions can be introduced to facilitate faster rates of language acquisition.

P47. Time course of reverberation adaptation when listening to natural sentences

Contractor Z A R (1), Culling J F (2), Walker K M M (1)

1. University of Oxford, Oxford, UK
2. Cardiff University, Cardiff, UK

Room reverberation provides important distance cues, but can also impair speech comprehension, especially for individuals with hearing loss. Previous research has demonstrated that listeners show improved speech intelligibility following adaptation to reverberation (Brandewie and Zahorik 2013, Beeston 2014). However, these studies have been limited to highly stereotyped sentence structures or isolated words. Here, we used the IEEE sentence corpus to explore the time course of reverberation adaption in more naturalistic sentences with low word predictability. Spoken sentences were presented over headphones in 5 simulated rooms (4 reverberant and 1 anechoic) in the presence of constant white noise. An adaptative staircase procedure was used to determine each listener's signal-to-noise threshold, and sentences were subsequently presented at threshold. Participants were required to report the final word of each presented sentence. Reverberation was introduced at varying time intervals prior to the target word in a given sentence to investigate the temporal dynamics of reverberation adaptation (following the approach of Beeston, 2014). We found that task performance improved in less reverberant rooms. Overall performance was also higher under binaural than monaural listening conditions, with reverberation impacting intelligibility more in monaural listening. Contrary to prior work, we did not observe an increase in word intelligibility with longer reverberation lead-in times, perhaps because the time course of reverberation adaptation is longer than the sentences presented here (~2.5s). These results show that reverberation impairs speech intelligibility in naturalistic sentences, particularly when binaural spatial cues are absent.

Brandewie & Zahorik, J Acoust Soc Am, 2013. DOI: 10.1121/1.4816263
Beeston et al., J Acoust Soc Am, 2014. DOI: 10.1121/1.4900596

P48. The effect of hearing aids on cognitive resources and the ease and effectiveness of communication: a randomised controlled trial

Loukieh S (1), Heinrich A (1), Schlittenlacher J (1), Urry E (2), Lewis P (3), Schmitt R (2), Kluk K (1)

1. The University of Manchester, Manchester, UK
2. Sonova AG, Zurich, Switzerland
3. Cardiff University, Cardiff, UK

Hearing aids improve audibility but their impact on listening effort and listening-related fatigue remains unclear. Cognitive ability and sleep may also affect these outcomes. This study aimed to determine whether first-time hearing aid use improves listening-related fatigue, listening effort, speech-in-noise comprehension, and wellbeing, and whether any effects depend on executive control or vary with age or sleep disturbance.

In a two-arm randomised controlled trial, 60 hearing aid–naive adults (≥60 years, hearing loss ≥25 dB HL) were randomised 1:1 to 4 weeks of hearing aid use or a control information session. Co-primary outcomes were listening-related fatigue and listening effort (both self-rated on 0–10 scales; effort assessed during a speech-in-noise task). Exploratory outcomes included discourse comprehension, sentence-in-noise recognition, executive control, wellbeing, and actigraphy-based sleep metrics. Data were analysed with ANCOVA (adjusting for baseline and demographics) and included planned mediation analysis.

Hearing aid users had significantly lower listening effort (p=0.019) and better speech-in-noise comprehension (p=0.011) than controls. No significant differences were seen in listening-related fatigue, sentence recognition, wellbeing, sleep, or executive control (all p>0.5). Outcomes did not differ by participant age or baseline sleep disturbance. Exploratory analysis indicated a trend towards greater benefits in those with more severe baseline hearing loss, though confirmation is needed. No mediation effects were found, and no adverse events were reported.

In older adults new to hearing aids, a 4-week hearing aid trial significantly reduced perceived listening effort and improved speech-in-noise comprehension, without significant effects on listening-related fatigue, sentence recognition, wellbeing, sleep, or executive control.